2025-26 S1 – BIO220 PROGRAM

Week	Lecture	Tutorials and Labs	Coursework
Week 1	Introduction to computational Biology.		
Week 2	Charges and Coulomb law. Electrostatic potential.		
Week 3	Atoms and molecules. The Schrodinger equation		
Week 4	Molecular mechanics. Force Fields.		Coursework 1 (in class test): Molecular Physics.
Week 5	Structure of proteins and nucleic acids.	Tutorial and Lab: Molecular visualization with VMD.	
Week 6	Structural models and protein folding. Homology modeling and Alphafold.	Tutorial and Lab: Homology modeling.	Coursework 2 (laboratory): Homology modeling and molecular visualization.
Week 7	Molecular surfaces and electrostatics.		
Week 8	Molecular docking.		Coursework 2 deadline
Week 9	Energy minimization. Normal modes analysis.	Tutorial and Lab: Molecular docking, normal modes.	
Week 10	Free energy and solvation.		
Week 11	Small molecules and chemical spaces. Chemoinformatics.	Tutorial and lab: Protein/Ligand modeling.	Coursework 3 (laboratory): Protein/Ligand modeling.
Week 12	Drug libraries, screening and selection.		
Week 13	Recap.		Coursework 3 deadline

STRUCTURE OF BIOLOGICAL MACROMOLECULES

Francesco Zonta, Ph.D.

Associate Professor

Department of Biosciences and Bioinformatics

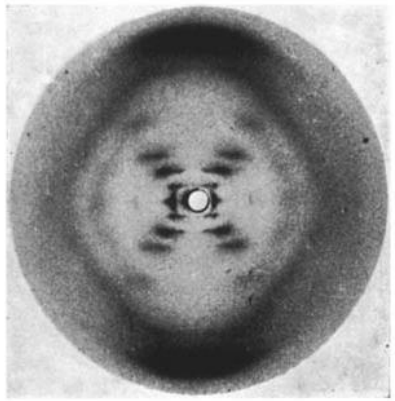
Room SB465, francesco.zonta@xjtlu.edu.cn



Xi'an Jiaotong-Liverpool University

西交利物浦大學

Watson, Crick, Gosling, Franklin and the Structure of DNA



Gosling, Franklin
Photo 51 (1952)



Watson and Crick (1953)

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[1] Young, F. B., Gerrard, H., and Jevons, W., *Phil. Mag.*, **40**, 149 (1920).
[2] Longuet-Higgins, M. S., *Mon. Not. Roy. Astro. Soc., Geophys. Supp.*, **5**, 285 (1949).
[3] Von Arx, W. S., Woods Hole Papers in Phys. Oceanog. Meteor., **11** (3) (1950).
[4] Ekman, V. W., *Arkiv. Mat. Astron. Fysik.* (*Stockholm*), **2** (11) (1905).

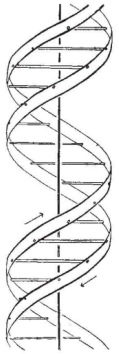
MOLECULAR STRUCTURE OF NUCLEIC ACIDS

A Structure for Deoxyribose Nucleic Acid

WE wish to suggest a structure for the salt of deoxyribose nucleic acid (D.N.A.). This structure has novel features which are of considerable biological interest.

A structure for nucleic acid has already been proposed by Pauling and Corey[1]. They kindly made their manuscript available to us in advance of publication. Their model consists of three intertwined chains, with the phosphates near the fibre axis, and the bases on the outside. In our opinion, this structure is unsatisfactory for two reasons: (1) We believe that the material which gives the X-ray diagrams is the salt, not the free acid. Without the acidic hydrogen atoms it is not clear what forces would hold the structure together, especially as the negatively charged phosphates near the axis will repel each other. (2) Some of the van der Waals distances appear to be too small.

Another three-chain structure has also been suggested by Fraser (in the press). In his model the phosphates are on the outside and the bases on the inside, linked together by hydrogen bonds. This structure as described is rather ill-defined, and for this reason we shall not comment on it.



This figure is purely diagrammatic. The two ribbons symbolize the two phosphate–sugar chains, and the horizontal rods the pairs of bases holding the chains together. The vertical line marks the fibre axis

We wish to put forward a radically different structure for the salt of deoxyribose nucleic acid. This structure has two helical chains each coiled round the same axis (see diagram). We have made the usual chemical assumptions, namely, that each chain consists of phosphate diester groups joining β-D-deoxyribofuranose residues with 3′,5′ linkages. The two chains (but not their bases) are related by a dyad perpendicular to the fibre axis. Both chains follow right-handed helices, but owing to the dyad the sequences of the atoms in the two chains run in opposite directions. Each chain loosely resembles Furberg's[2] model No. 1; that is, the bases are on the inside of the helix and the phosphates on the outside. The configuration of the sugar and the atoms near it is close to Furberg's 'standard configuration', the sugar being roughly perpendicular to the attached base. There is a residue on each chain every 3·4 A. in the z-direction. We have assumed an angle of 36° between adjacent residues in the same chain, so that the structure repeats after 10 residues on each chain, that is, after 34 A. The distance of a phosphorus atom from the fibre axis is 10 A. As the phosphates are on the outside, cations have easy access to them.

The structure is an open one, and its water content is rather high. At lower water contents we would expect the bases to tilt so that the structure could become more compact.

The novel feature of the structure is the manner in which the two chains are held together by the purine and pyrimidine bases. The planes of the bases are perpendicular to the fibre axis. They are joined together in pairs, a single base from one chain being hydrogen-bonded to a single base from the other chain, so that the two lie side by side with identical z-co-ordinates. One of the pair must be a purine and the other a pyrimidine for bonding to occur. The hydrogen bonds are made as follows: purine position 1 to pyrimidine position 1; purine position 6 to pyrimidine position 6.

If it is assumed that the bases only occur in the structure in the most plausible tautomeric forms (that is, with the keto rather than the enol configurations) it is found that only specific pairs of bases can bond together. These pairs are: adenine (purine) with thymine (pyrimidine), and guanine (purine) with cytosine (pyrimidine).

In other words, if an adenine forms one member of a pair, on either chain, then on these assumptions the other member must be thymine; similarly for guanine and cytosine. The sequence of bases on a single chain does not appear to be restricted in any way. However, if only specific pairs of bases can be formed, it follows that if the sequence of bases on one chain is given, then the sequence on the other chain is automatically determined.

It has been found experimentally[3,4] that the ratio of the amounts of adenine to thymine, and the ratio of guanine to cytosine, are always very close to unity for deoxyribose nucleic acid.

It is probably impossible to build this structure with a ribose sugar in place of the deoxyribose, as the extra oxygen atom would make too close a van der Waals contact.

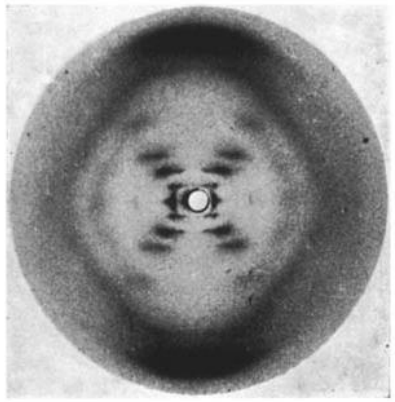
The previously published X-ray data[5,6] on deoxyribose nucleic acid are insufficient for a rigorous test of our structure. So far as we can tell, it is roughly compatible with the experimental data, but it must be regarded as unproved until it has been checked against more exact results. Some of these are given in the following communications. We were not aware of the details of the results presented there when we devised our structure, which rests mainly though not entirely on published experimental data and stereochemical arguments.

It has not escaped our notice that the specific pairing we have postulated immediately suggests a possible copying mechanism for the genetic material.

Full details of the structure, including the conditions assumed in building it, together with a set of co-ordinates for the atoms, will be published elsewhere.

We are much indebted to Dr. Jerry Donohue for constant advice and criticism, especially on interatomic distances. We have also been stimulated by a knowledge of the general nature of the unpublished experimental results and ideas of Dr. M. H. F. Wilkins, Dr. R. E. Franklin and their co-workers at

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No. 4356 April 25, 1953 NATURE 737

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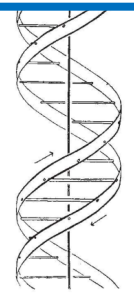
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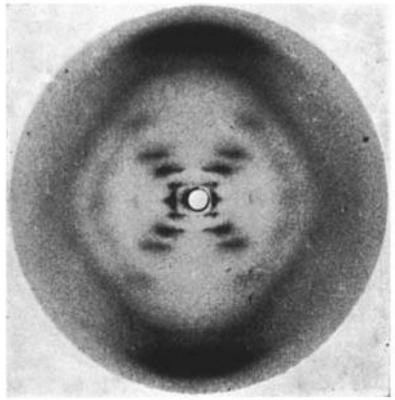
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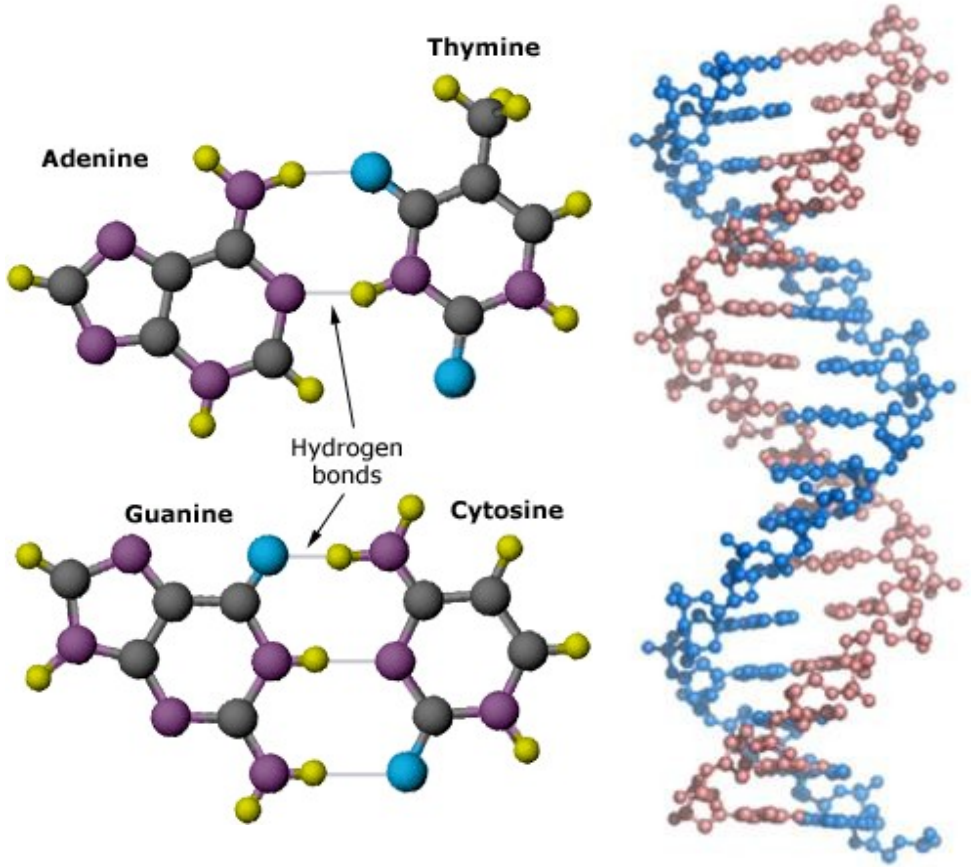
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Perutz, Kendrew and Protein Structures





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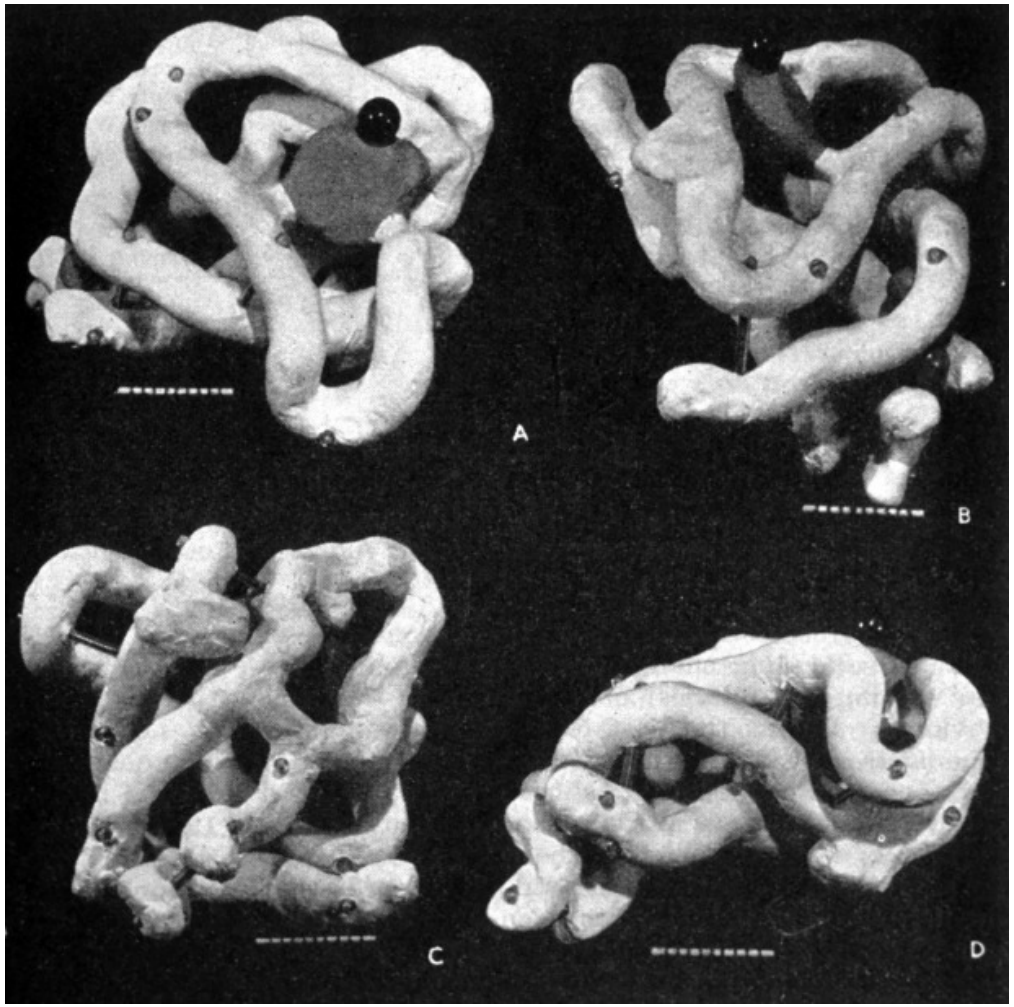


Molecular modeling in the 60's looks like lots of fun!

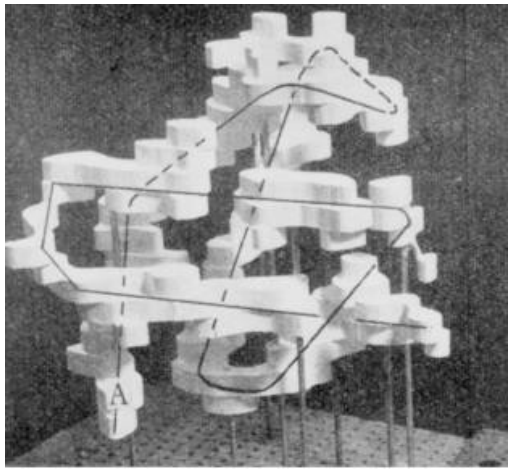


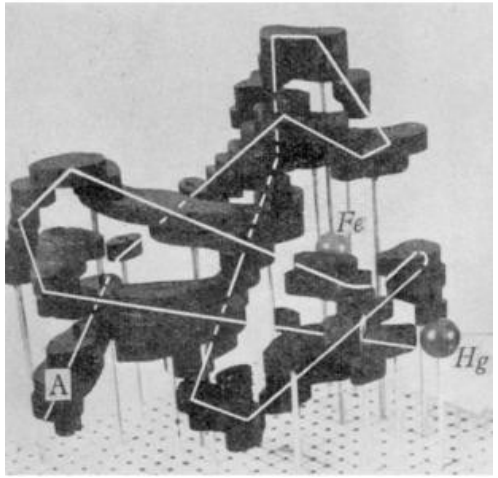
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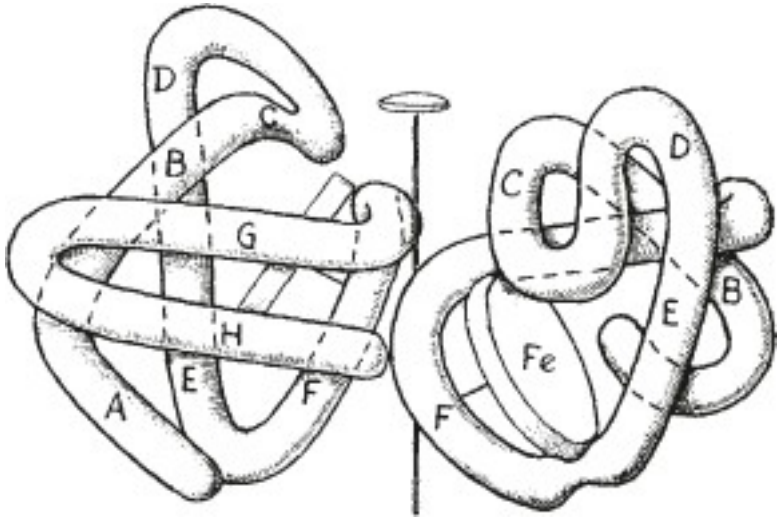
Perutz, Kendrew and Protein Structures



Kendrew (1958) structure of myoglobin







Perutz (1960) structure of hemoglobin

Roger Kornberg and the RNA Polymerase II

Structural Basis of Transcription: RNA Polymerase II at 2.8 Ångstrom Resolution

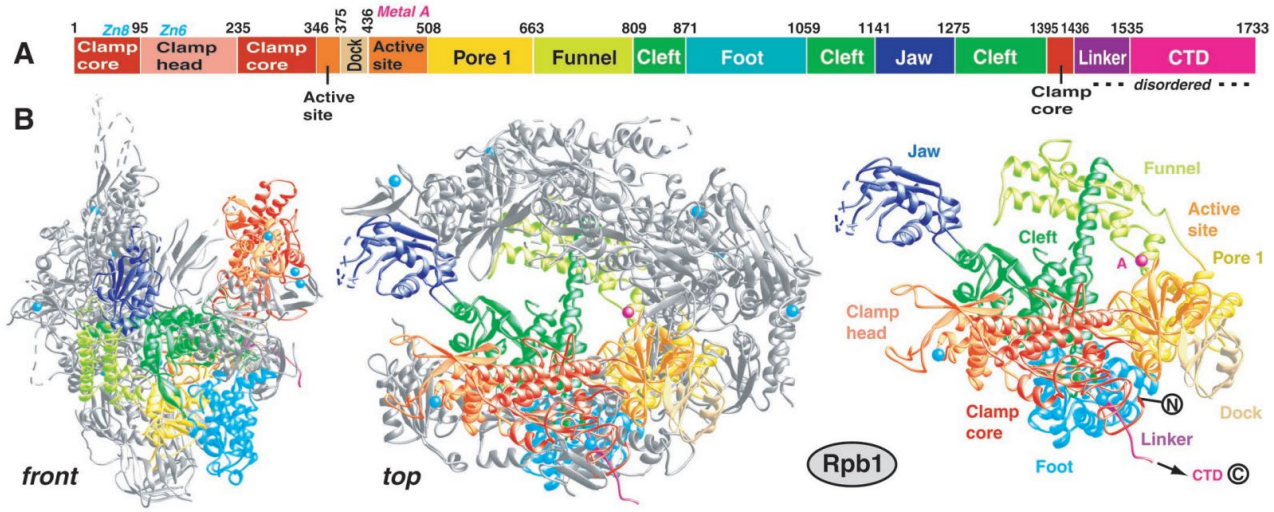
PATRICK CRAMER, DAVID A. BUSHNELL, AND ROGER D. KORNBERG Authors Info & Affiliations

SCIENCE · 19 Apr 2001 · Vol 292, Issue 5523 · pp. 1863-1876 · DOI: 10.1126/science.1059493

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Abstract

Structures of a 10-subunit yeast RNA polymerase II have been derived from two crystal forms at 2.8 and 3.1 angstrom resolution. Comparison of the structures reveals a division of the polymerase into four mobile modules, including a clamp, shown previously to swing over the active center. In the 2.8 angstrom structure, the clamp is in an open state, allowing entry of straight promoter DNA for the initiation of transcription. Three loops extending from the clamp may play roles in RNA unwinding and DNA rewinding during transcription. A 2.8 angstrom difference Fourier map reveals two metal ions at the active site, one persistently bound and the other possibly exchangeable during RNA synthesis. The results also provide evidence for RNA exit in the vicinity of the carboxyl-terminal repeat domain, coupling synthesis to RNA processing by enzymes bound to this domain.





2006 Nobel Prize in Chemistry



Roger D. Kornberg

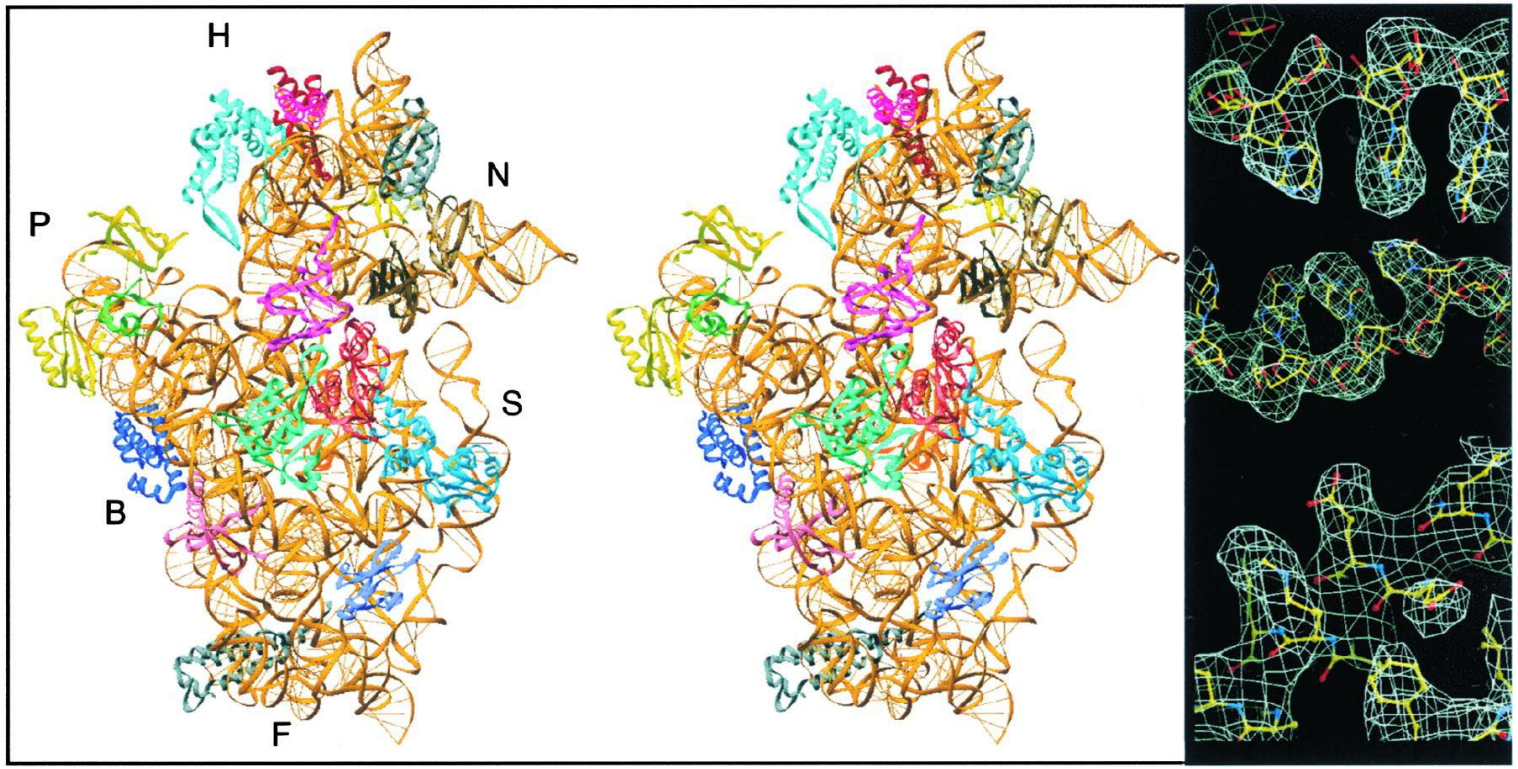
"for his studies of the molecular basis of eukaryotic transcription"

Yonath, Ramakrishnan, Steitz and the Ribosome

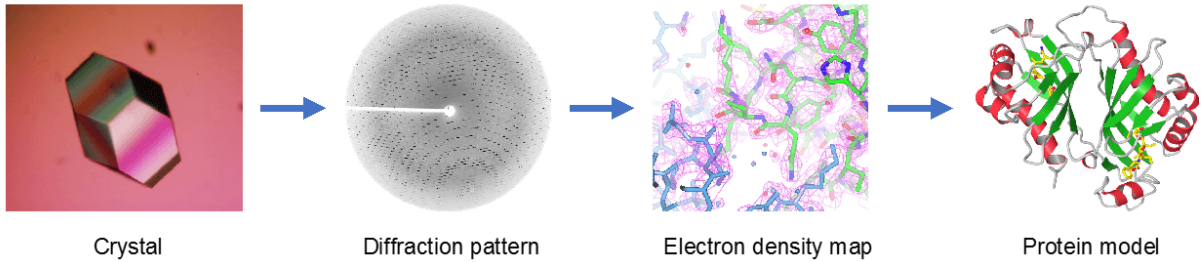
Cell, Vol. 102, 615–623, September 1, 2000, Copyright ©2000 by Cell Press

Structure of Functionally Activated Small Ribosomal Subunit at 3.3 Å Resolution

Frank Schluenzen,*‖ Ante Tocilj,*‖ Raz Zarivach,†
Joerg Harms,* Marco Gluehmann,*
Daniela Janell,* Anat Bashan,† Heike Bartels,†
Ilana Agmon,† François Franceschi,‡§
and Ada Yonath*†
*Max-Planck-Research Unit for Ribosomal Structure
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Germany
†Weizmann Institute
76100 Rehovot
Israel
‡Max-Planck-Institute for Molecular Genetics
14195 Berlin
Germany



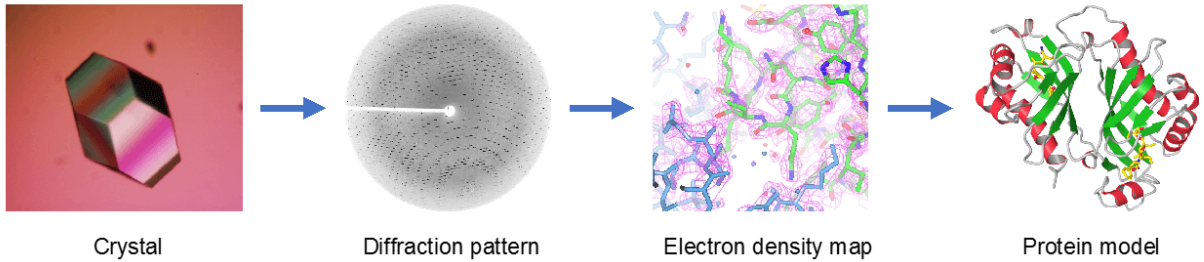
Experimental methods



X-Ray Crystallography

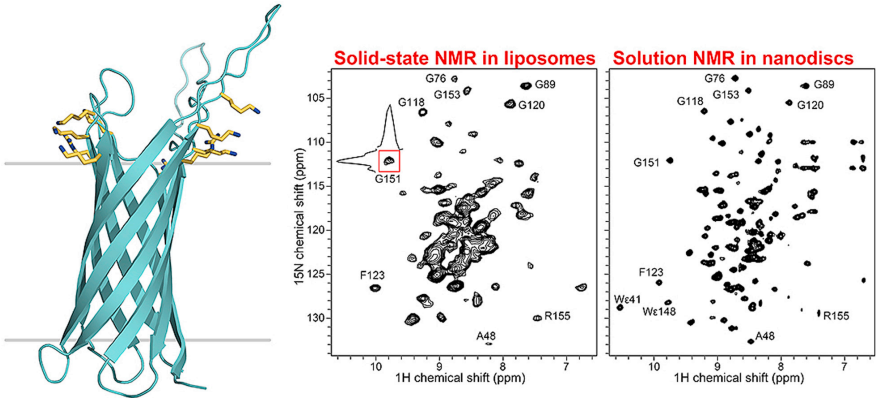
- Requires crystallizing the protein of interest
- X-rays are diffracted by the repeating array of molecules in the crystal
- Diffraction pattern allows 3D structures to be determined at atomic resolution
- Established method but challenges in crystallization

Experimental methods



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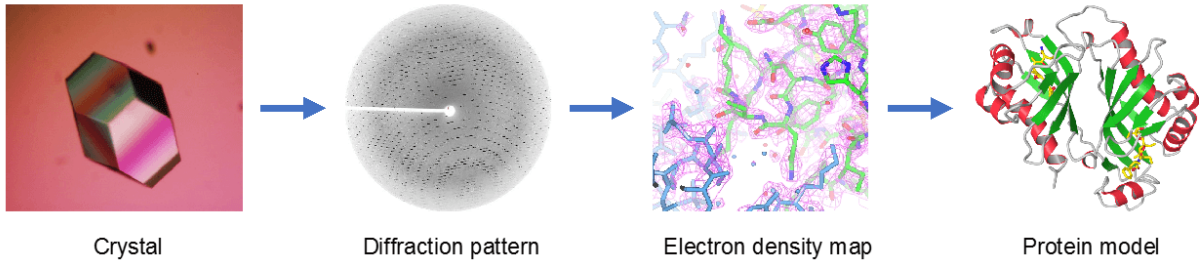
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NMR Spectroscopy

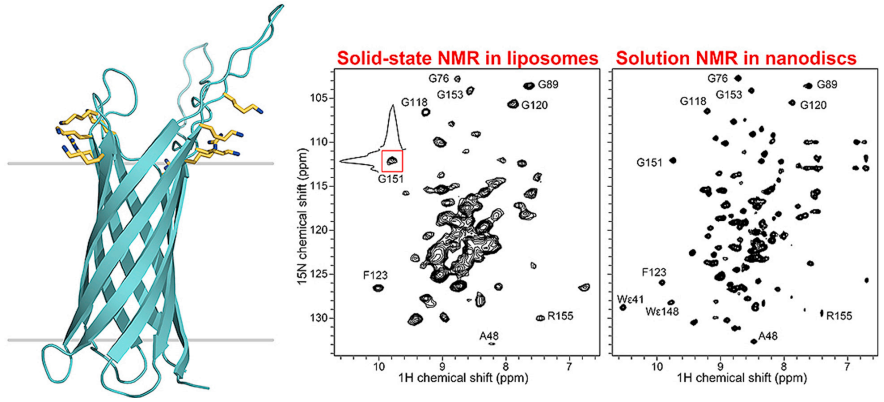
- Measures magnetic properties of atomic nuclei in a protein solution
- Provides info on atoms' chemical environment and dynamics
- Calculates 3D structure from nuclear distances/angles
- Can study proteins in near-native solution conditions
- Size limits of ~50-100 kDa for traditional NMR

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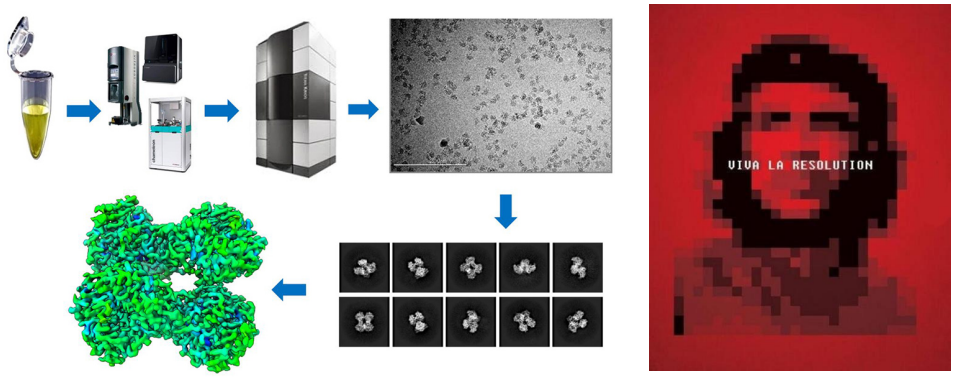
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Cryo-Electron Microscopy (Cryo-EM/ET)

- Rapidly freezes protein samples in vitreous ice
- Electrons scattered by frozen biomolecules recorded
- 3D maps reconstructed from 2D projection images
- Recent resolution "revolution" reaching near-atomic levels
- Can study large complexes, membrane proteins, dynamics

Structural biology in 2020s: large complexes and viruses

Architecture of African swine fever virus and implications for viral assembly

NAN WANG, DONGMING ZHAO, JIALING WANG, YANGLING ZHANG, MING WANG, YAN GAO, FANG LI, JINGFEI WANG, ZHIGAO BU, ZIHE RAO, AND XIANGXI WANG fewer Authors Info & Affiliations

SCIENCE • 17 Oct 2019 • Vol 366, Issue 6465 • pp. 640-644 • DOI: 10.1126/science.aaz1439

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Unveiling African swine fever virus

African swine fever virus (ASFV) is highly contagious and often lethal. With no vaccine or effective treatment, infections often require large-scale culling of pigs. Wang *et al.* apply cutting-edge cryo–electron microscopy techniques to determine the structure of this very large DNA virus. An 8.8-angstrom-resolution reconstruction shows the five layers of the virus, and the fourth capsid layer could be reconstructed at 4.8-angstrom resolution. The structure reveals epitopes in the major capsid protein that distinguish ASFV from other nucleocytoplasmic large DNA viruses and shows how the minor capsid proteins stabilize the capsid.

Science, this issue p. 640

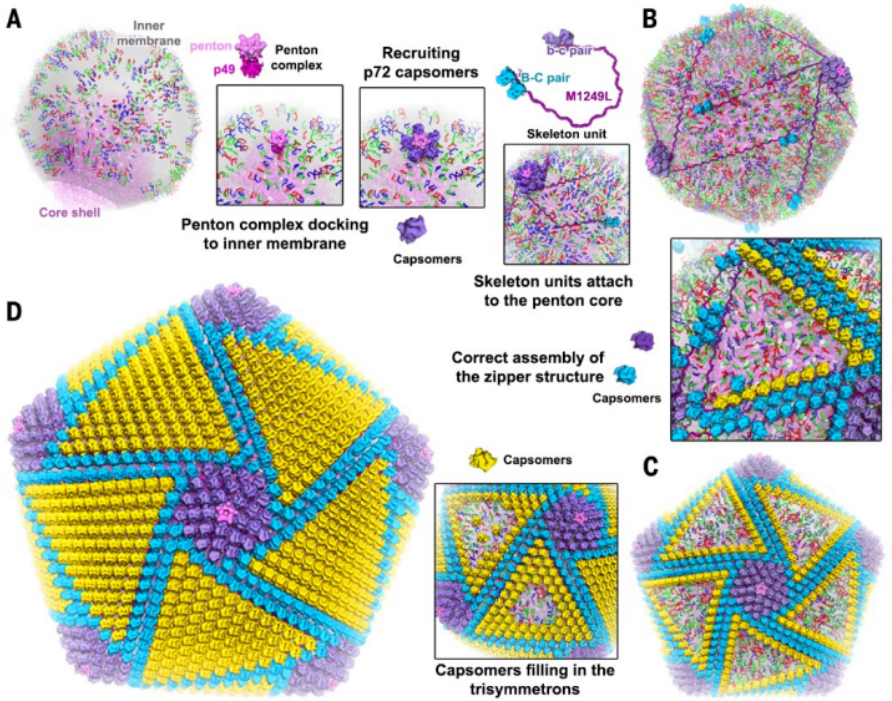
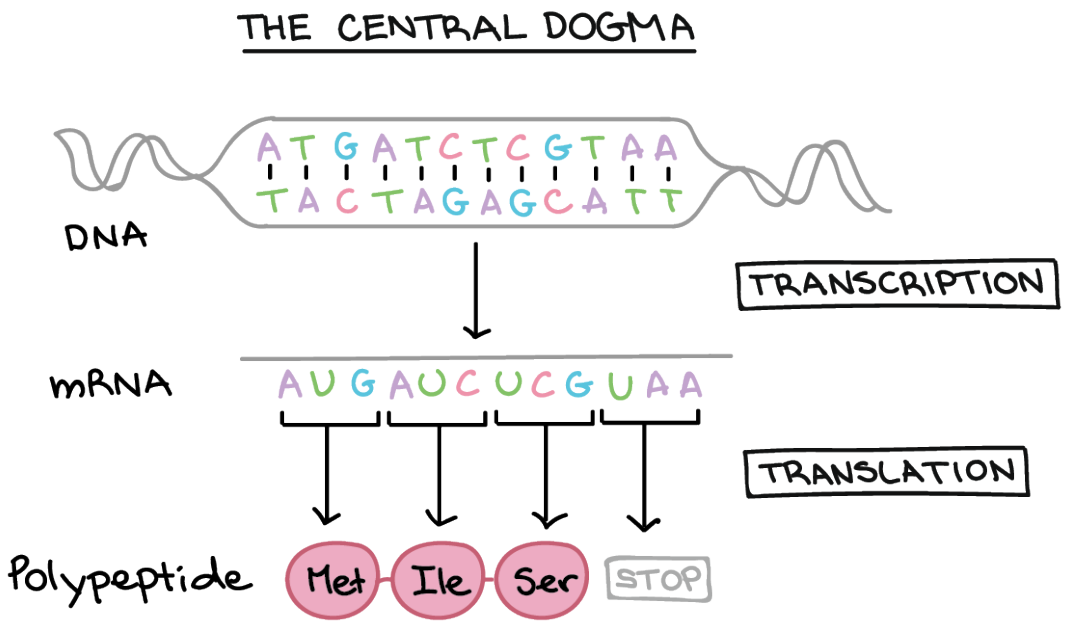


Fig. 4. The proposed assembly pathway for the ASFV capsid. (**A**) Formation of the penton cores on the inner membrane. The ASFV capsid assembly begins with appearance of viral inner membrane precursors that envelop the core shell and nucleoid progressively. At the early stage, the inner membranes exhibit generally open structures, with various inner membrane proteins, e.g., floating on p17. The penton complexes associate with the inner membrane, then recruit p72 capsomers to form the penton cores, initiating the assembly. (**B**) Correct assembly of the zipper structures mediated by skeleton protein M1249L, p17 and p72 capsomers. (**C**) Construction of polyhedral cage by twelve pentasymmetrons and 30 zippers. (**D**) Accompanying the formation of the polyhedral framework, p72 capsomers fill in the trisymmetrons to complete the capsid assembly.

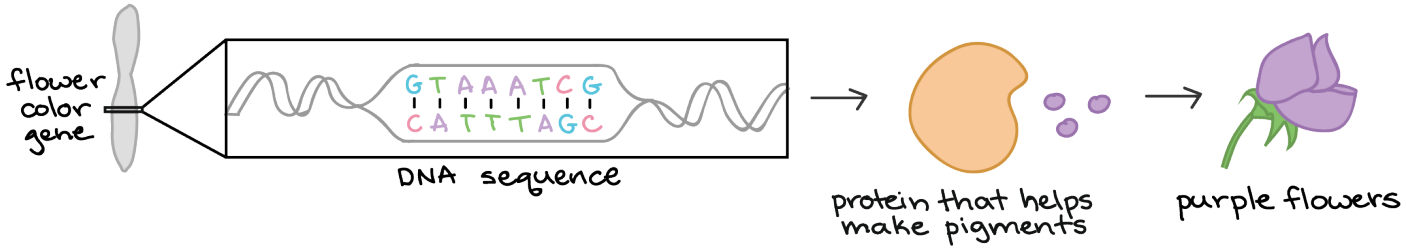
Nobel laureates and key discoveries in structural biology

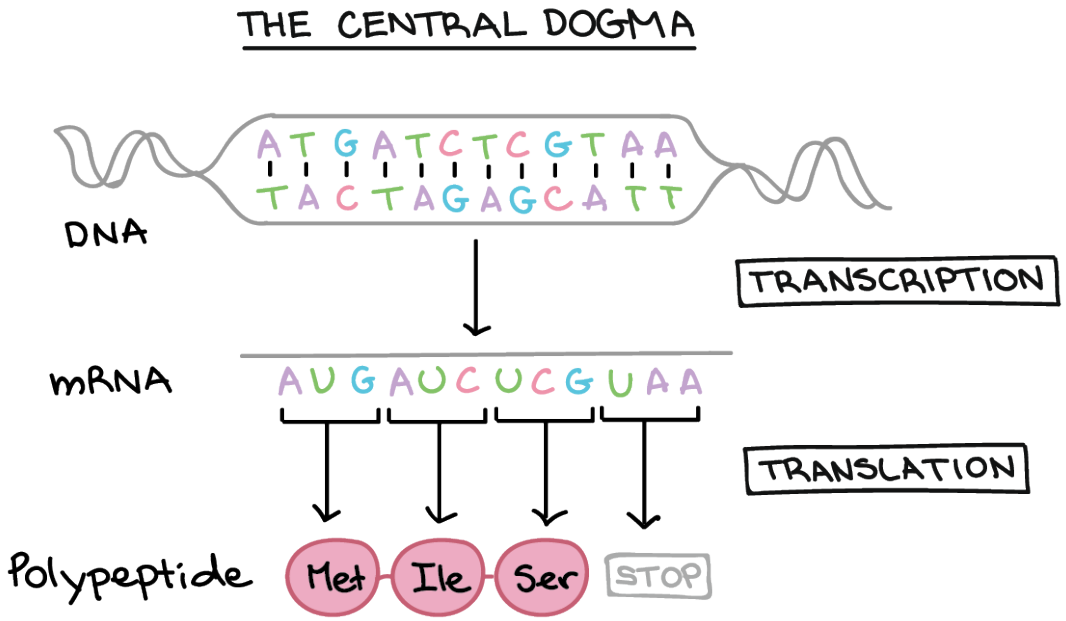
Year	Scientist(s)	Key Discovery
1954	Linus Pauling	Discovered α-helices and β-sheets as fundamental protein structures.
1962	Francis Crick, James Watson, Maurice Wilkins	Solved the double helix structure of DNA (based on Rosalind Franklin's X-ray data).
1962	Max Perutz, John Kendrew	Solved the first protein structures (hemoglobin and myoglobin) using X-ray crystallography.
1969	Derek Barton, Odd Hassel	Developed the concept of molecular conformation, crucial for understanding protein folding.
1985	Herbert Hauptman, Jerome Karle	Developed direct methods for solving crystal structures, improving X-ray crystallography.
1988	Johann Deisenhofer, Robert Huber, Hartmut Michel	Solved the first membrane protein structure (photosynthetic reaction center) using X-ray crystallography.
1997	Paul Boyer, John Walker, Jens Skou	Determined the mechanism of ATP synthase, showing its rotary function.
2002	Kurt Wüthrich	Developed NMR spectroscopy for determining protein structures in solution.
2003	Roderick MacKinnon	Solved the K^+ ion channel structure, revealing ion transport mechanisms.
2006	Roger Kornberg	Solved the RNA polymerase structure, revealing the molecular basis of transcription.
2009	Venkatraman Ramakrishnan, Thomas Steitz, Ada Yonath	Solved the ribosome structure, explaining protein synthesis at atomic resolution.
2013	Martin Karplus, Michael Levitt, Arieh Warshel	Developed molecular dynamics simulations for protein structure modeling.
2017	Jacques Dubochet, Joachim Frank, Richard Henderson	Developed cryo-electron microscopy (Cryo-EM), enabling high-resolution structure determination of biomolecules.
2021	David Julius, Ardem Patapoutian	Structural studies of temperature and pain receptors (TRP ion channels).
2023	Moungi Bawendi, Louis Brus, Alexei Ekimov	Developed quantum dots, relevant for biomolecular imaging.
2024	David Baker, Demis Hassabis, John Jumper	David Baker: Pioneered computational protein design, creating novel proteins with specific functions. Demis Hassabis and John Jumper: Developed AlphaFold, an AI model that accurately predicts protein structures, revolutionizing structural biology.

The Central Dogma in Biology



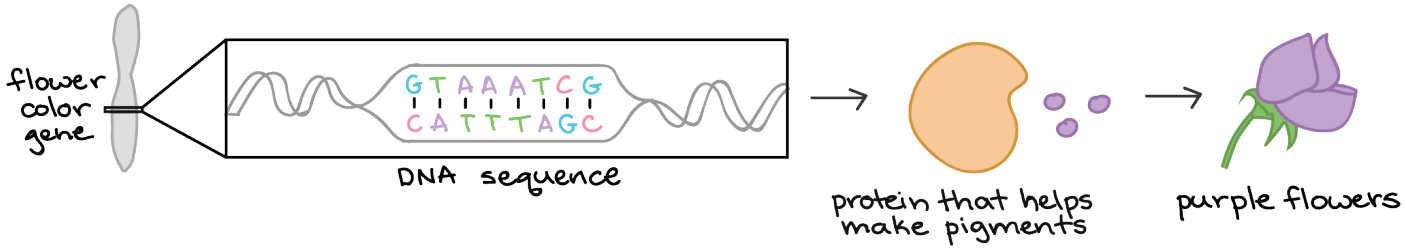
The Central Dogma in Biology

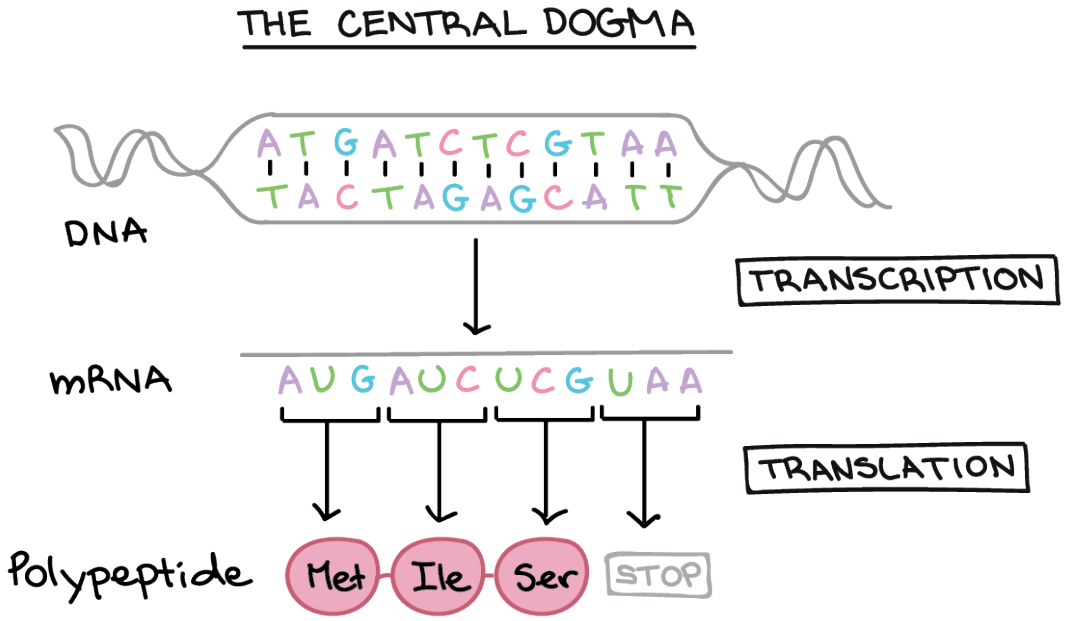


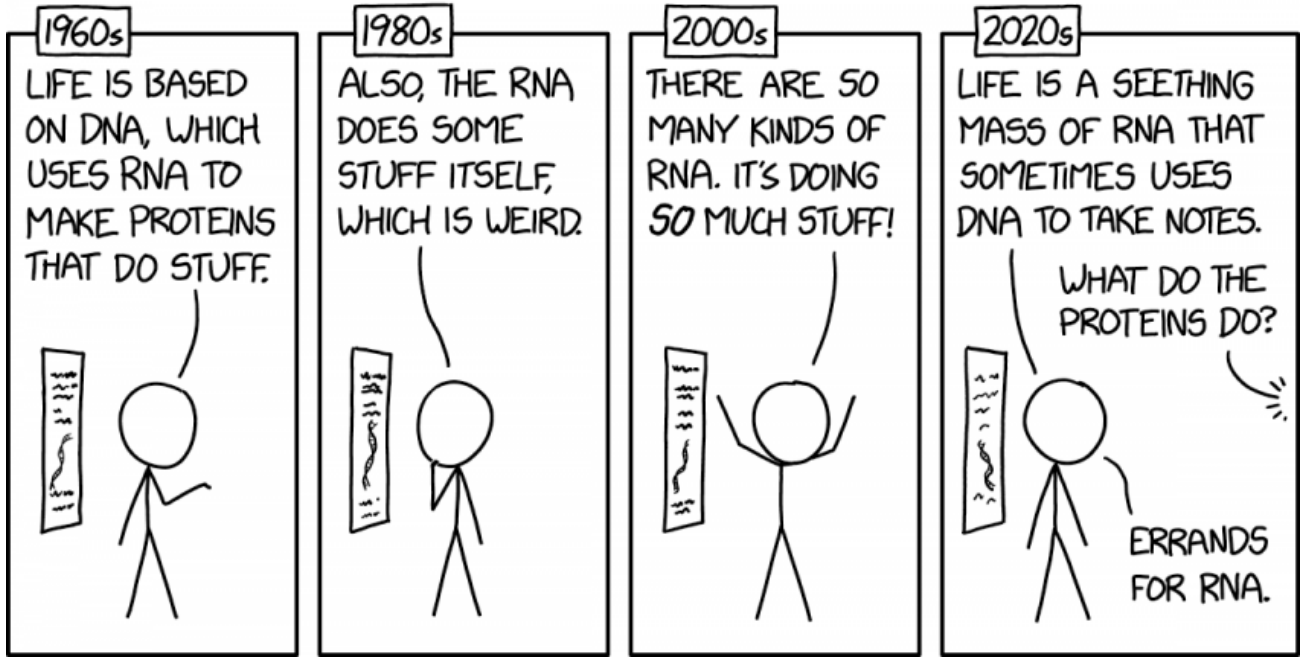


Francesco Zonta – Room SB465, Department of Biosciences and Bioinformatics – francesco.zonta@xjtlu.edu.cn

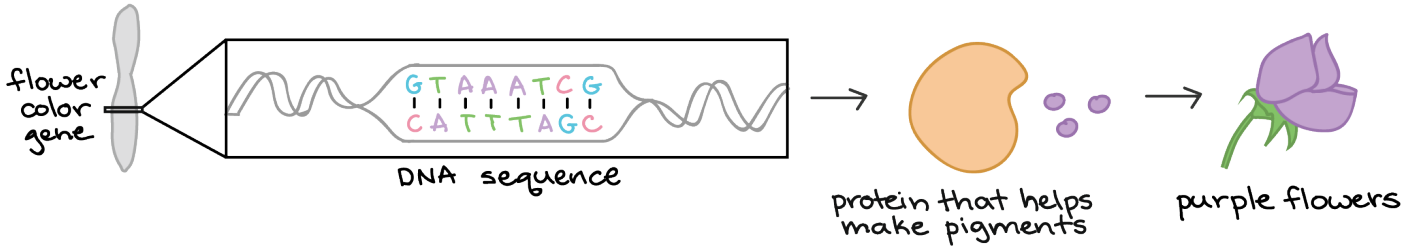
The Central Dogma in Biology

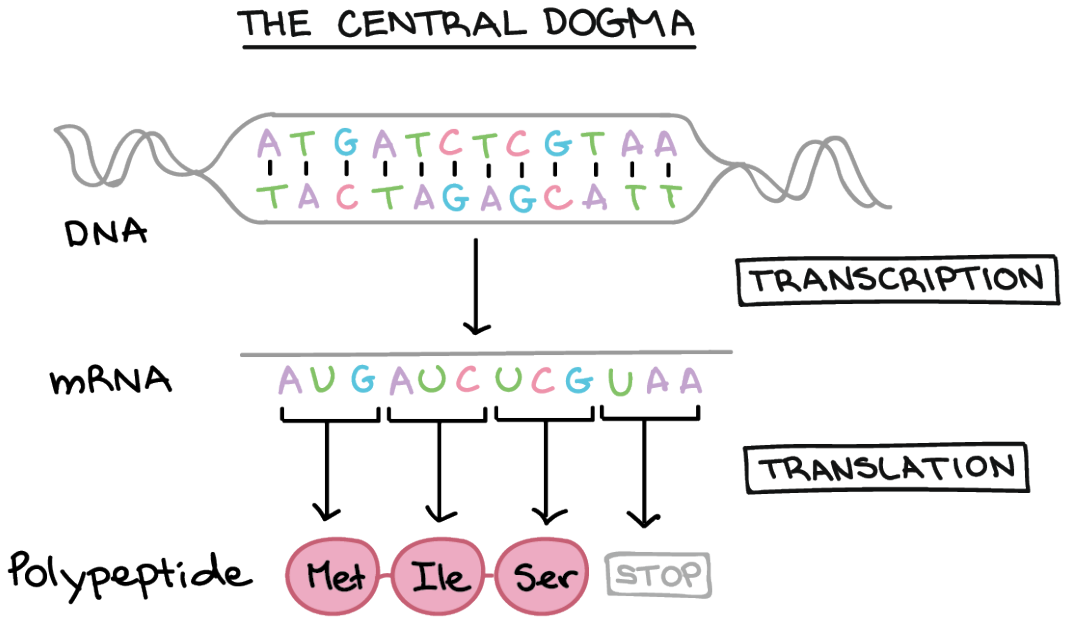


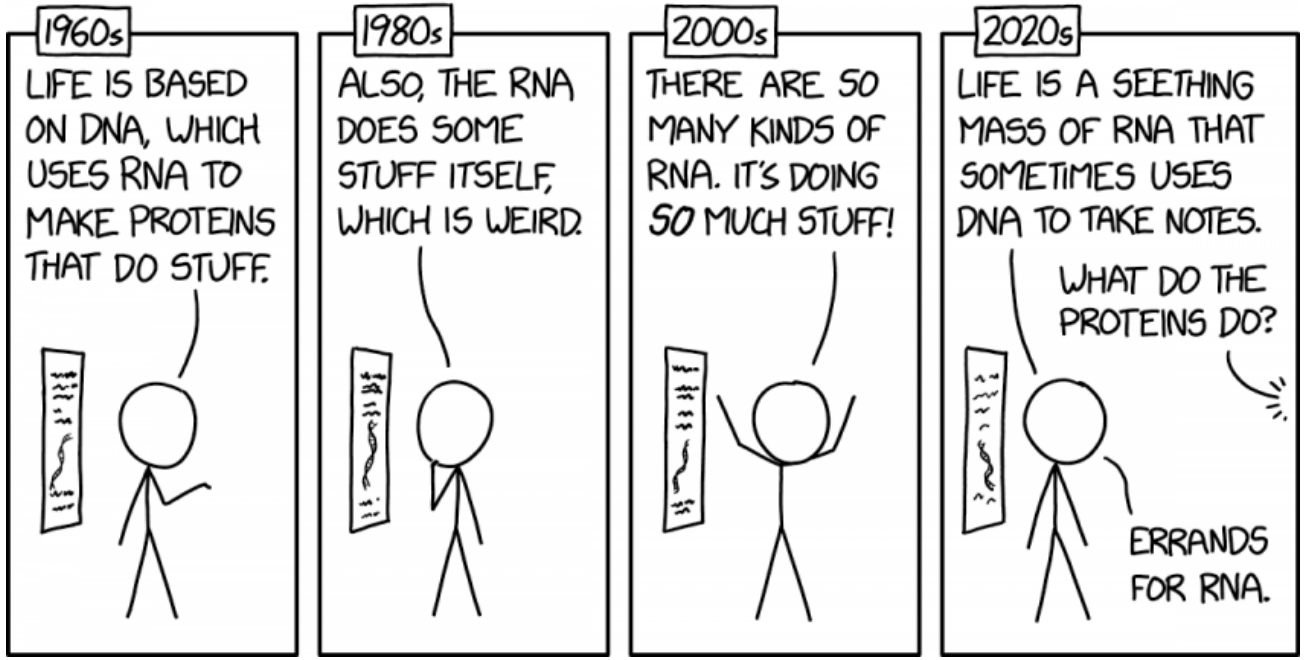




The Central Dogma in Biology

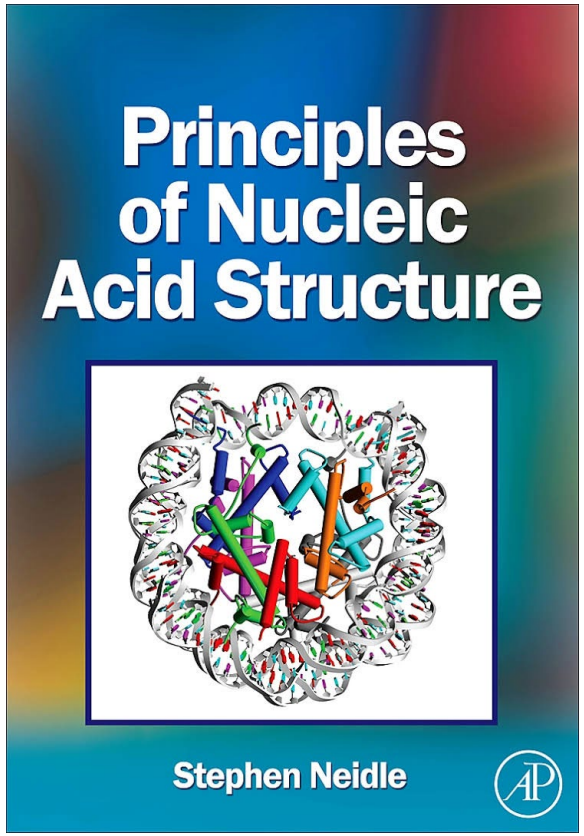


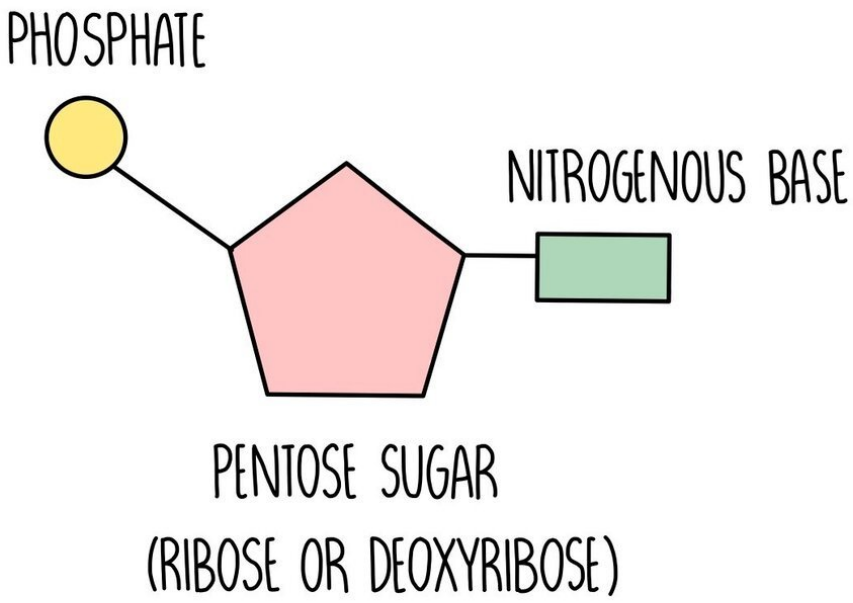




"DNA makes RNA, RNA makes proteins, proteins make money" (R. Lerner)

Nucleic Acids Structure





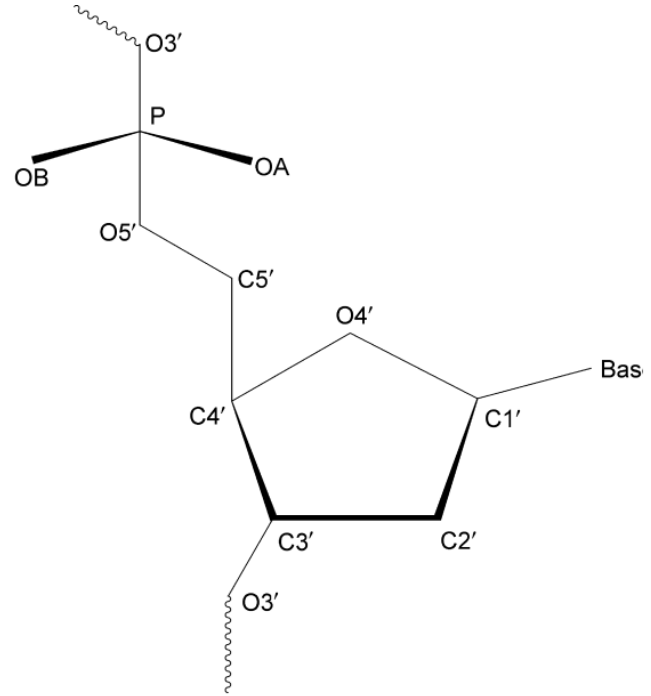
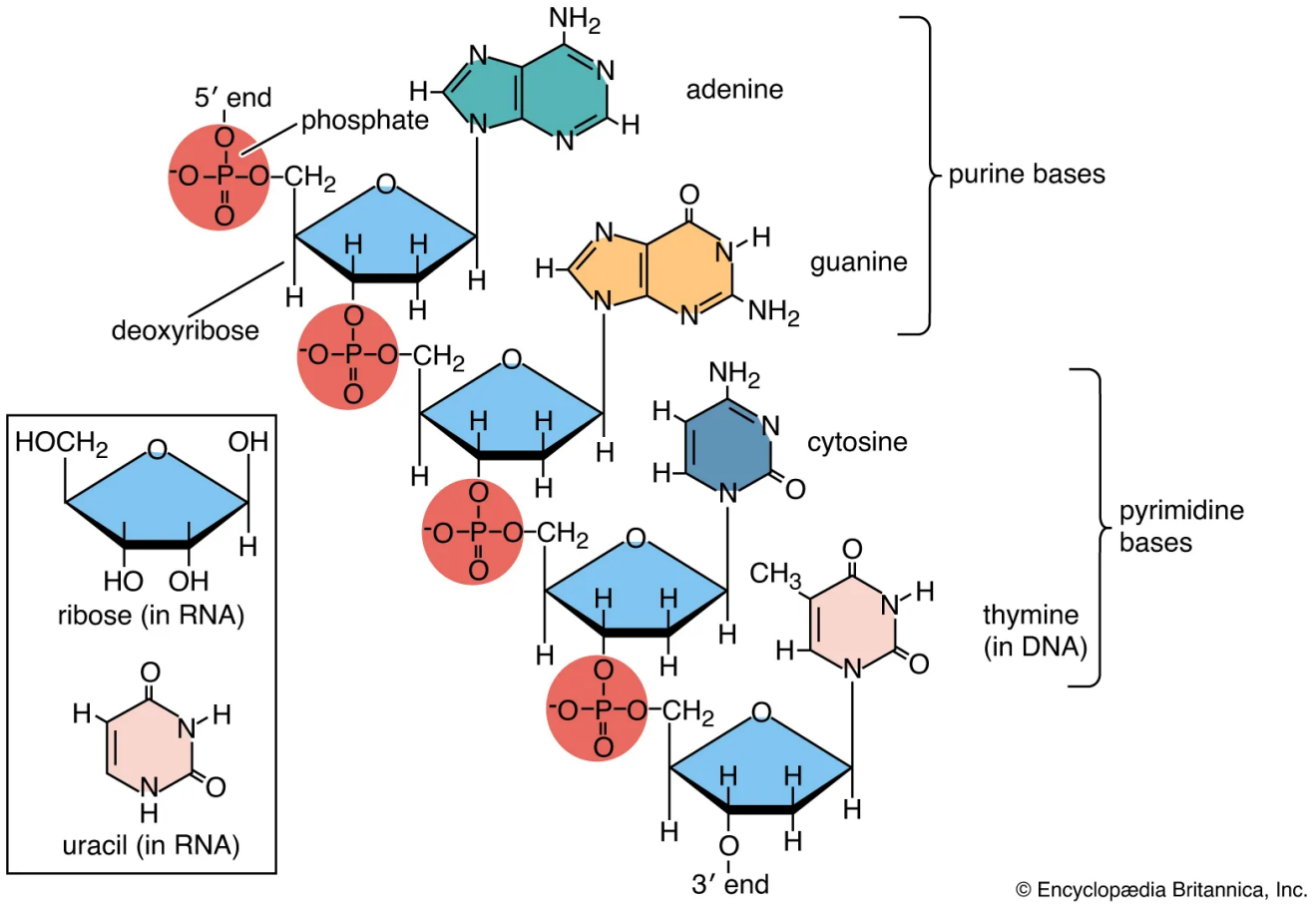


Figure 2.2 The organization of repeating units in a polynucleotide chain.

DNA and RNA are polymers of nucleotides (polynucleotides)

Polymerization and pairing of nucleotides



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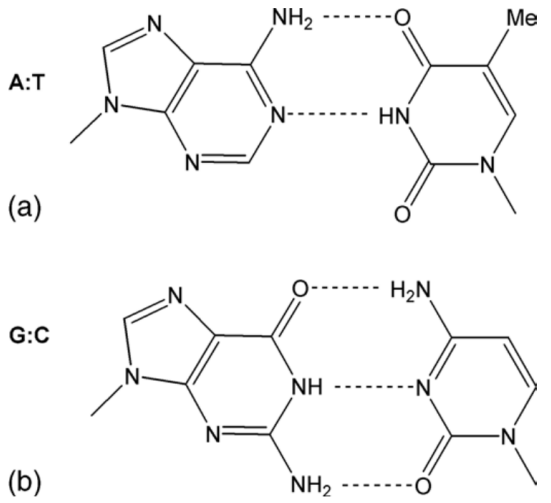


Figure 2.4 (a) A•T and (b) G•C base pairs, showing Watson–Crick hydrogen bonding.

Table 2.1 Hydrogen-Bond Distances (in Å) in Watson–Crick Base Pairs (Seeman *et al.*, 1976; Rosenberg *et al.*, 1976), in the Crystalline State. Estimated Standard Deviations are in Parentheses

U•A:	N3-H.....N1	2.835(8)
	O4.....H-N6	2.940(8)
C•G:	O2.....H-N2	2.86(1)
	N3.....H-N1	2.95(1)
	N4-H.....O6	2.91(1)

Base pair flexibility

Bases from opposed strands do not always pair in standard way. Local changes are very common.

For individual base pairs:

(a) Propeller twist (ω)
(b) Buckle (κ)
(c) Inclination (η)

For base pair steps:

(e) Helical twist (Ω)
(f) Roll (ρ)
(g) Slide

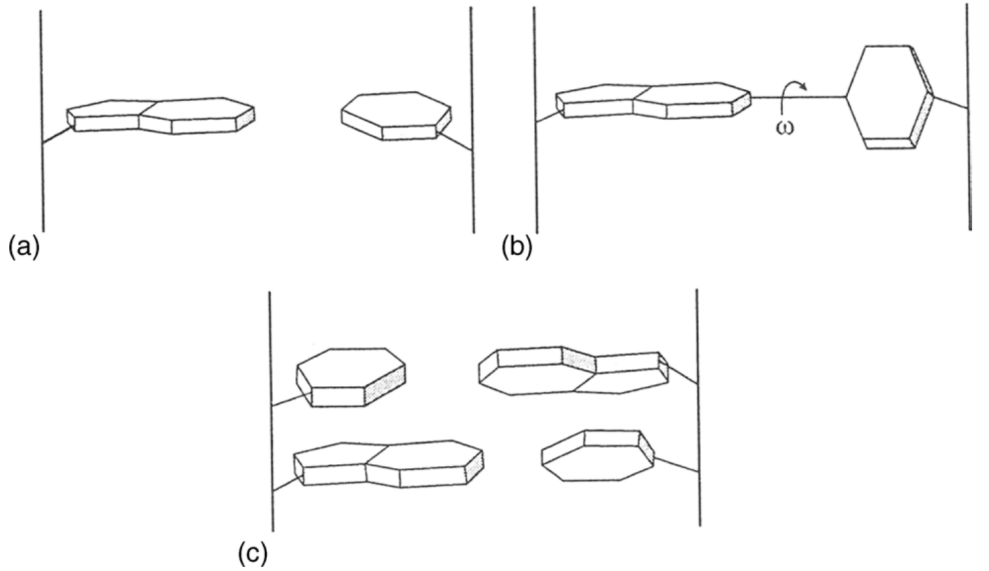


Figure 2.5 Schematic views of a base pair with (a) zero and (b) high propeller twist. Panel (c) shows the effects of propeller twist on two successive base pairs.

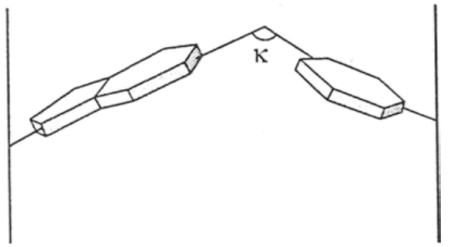
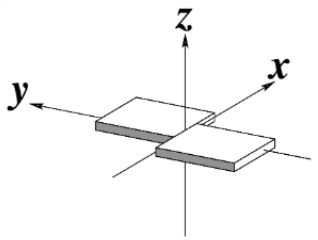


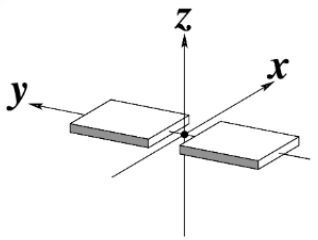
Figure 2.6 A schematic view of base-pair buckle.

Base pair flexibility

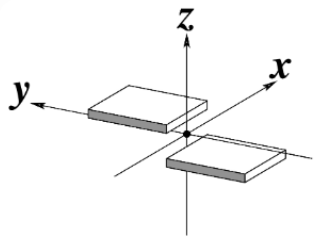
Shear (*Sx*)



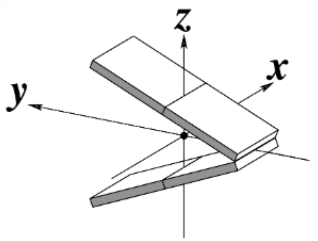
Stretch (*Sy*)



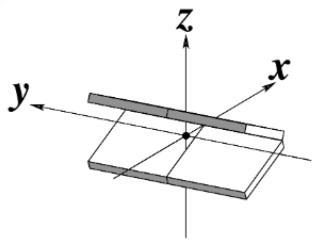
Stagger (*Sz*)



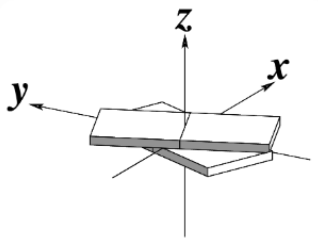
Tilt (*τ*)



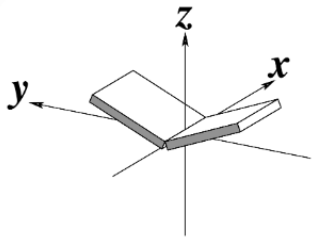
Roll (*ρ*)



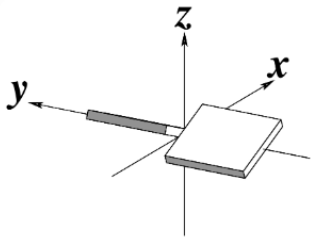
Twist (*ω*)



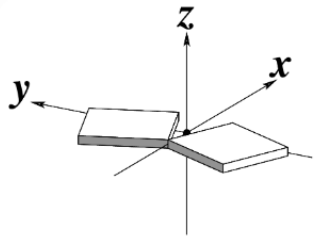
Buckle (*κ*)



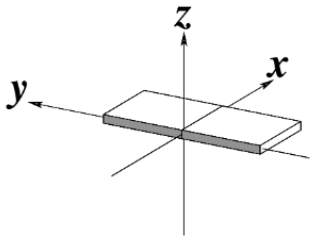
Propeller (*π*)



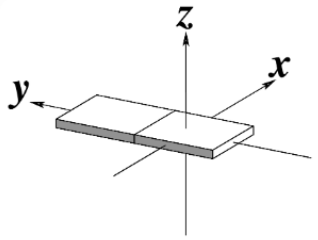
Opening (*σ*)



***x*-displacement (*dx*)**



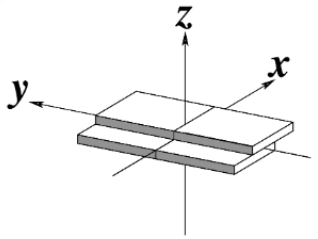
***y*-displacement (*dy*)**



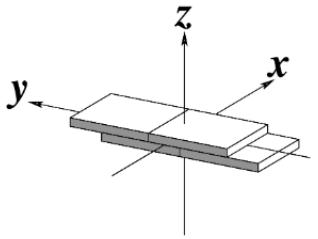
Helical Rise (*h*)

As for Rise above
(for illustration purpose)

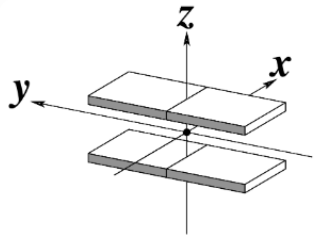
Shift (*Dx*)



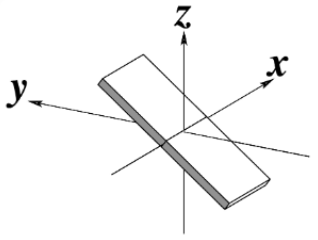
Slide (*Dy*)



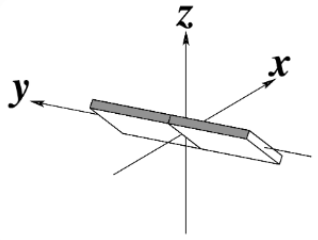
Rise (*Dz*)



Inclination (*η*)



Tip (*θ*)



Helical Twist (*Ωh*)

As for Twist above
(for illustration purpose)

Sugar Puckers

The five-membered deoxyribose sugar ring in DNA is inherently nonplanar. This nonplanarity is termed puckering.

The ring puckering arises from the effect of nonbonded interactions between substituents at the four ring carbon atoms – the energetically most stable conformation for the ring has all substituents as far apart as possible.

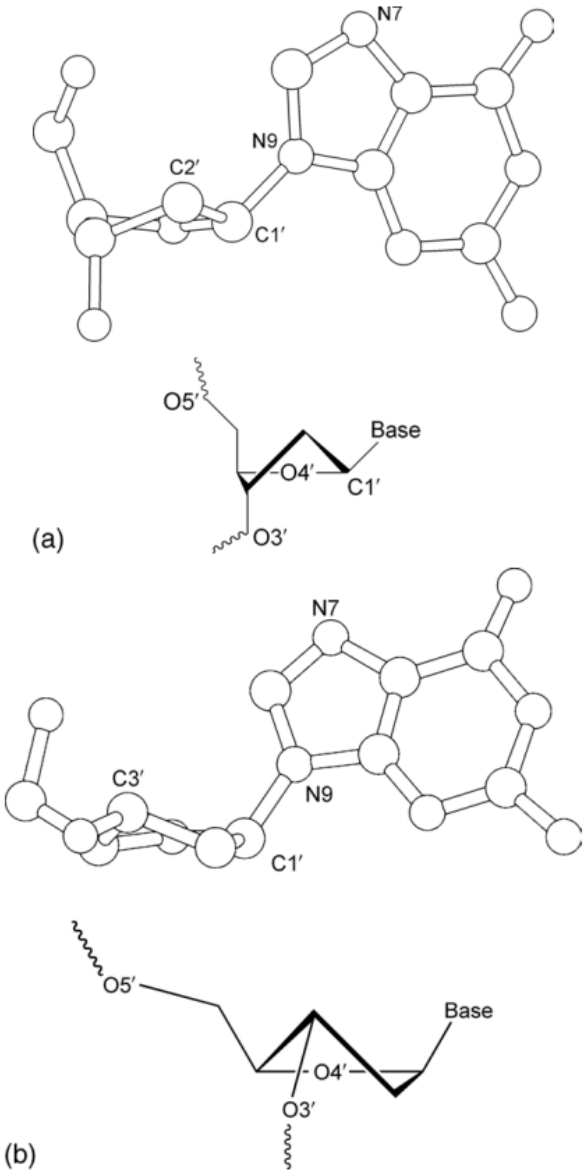


Figure 2.12 (a) C2′-*endo* sugar puckering for a guanosine nucleoside, viewed along the plane of the sugar ring. (b) C3′-*endo* sugar puckering for a guanosine nucleoside, with the sugar viewed in the same direction as in panel (a).

Torsional angles

The glycosidic bond links a deoxyribose sugar and a base. The torsion angle χ around this single bond can in principle adopt a wide range of values. Theory has predicted two principal low-energy domains for the glycosidic angle.

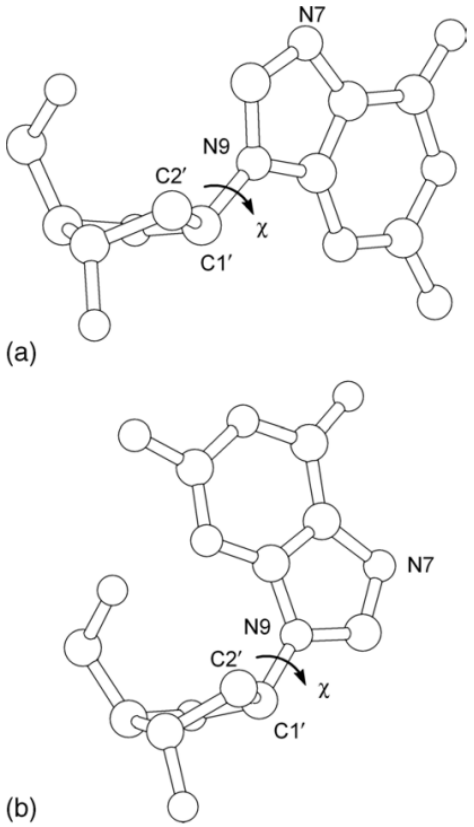


Figure 2.13 (a) A guanosine nucleoside with the glycosidic angle χ set in an *anti* conformation. (b) A guanosine nucleoside in the *syn* conformation.

The phosphodiester backbone of an oligonucleotide has six variable torsion angles designated α.... ζ

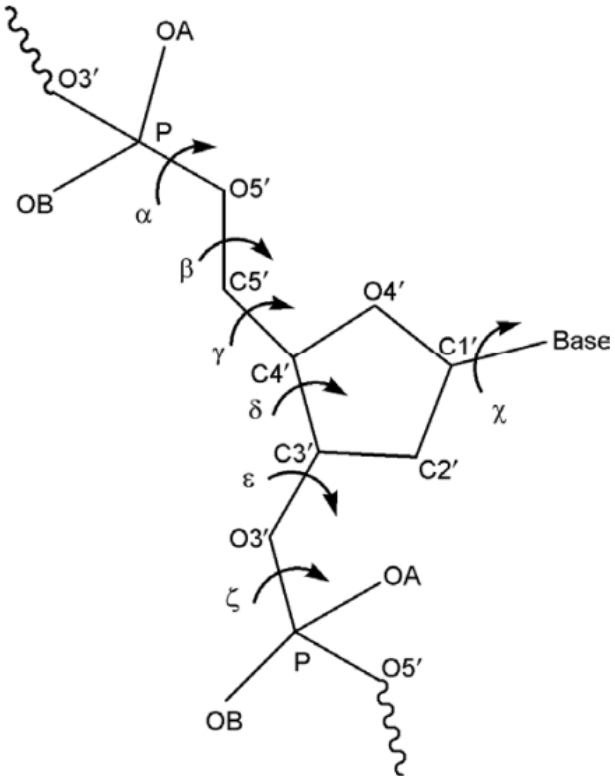
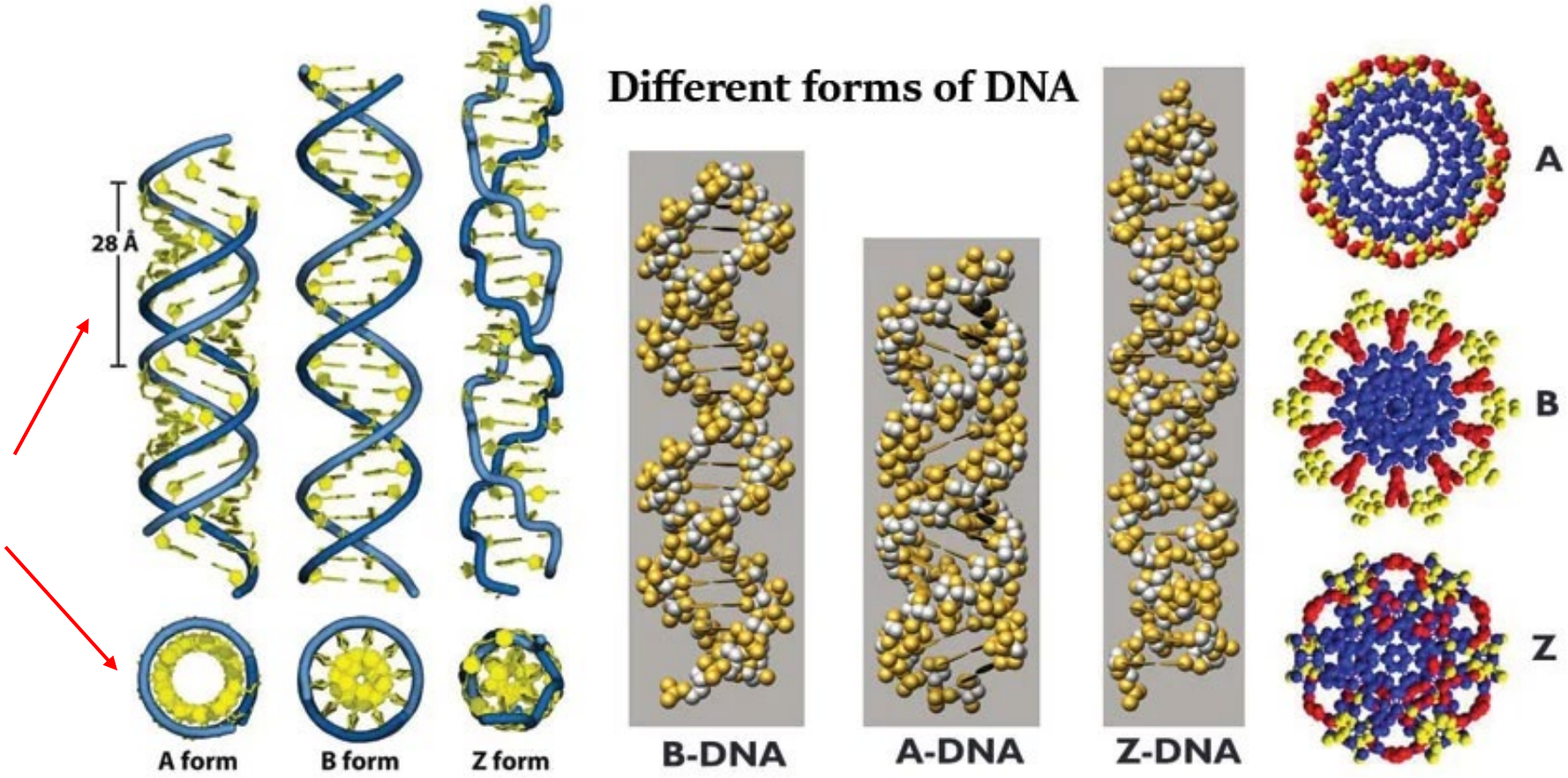


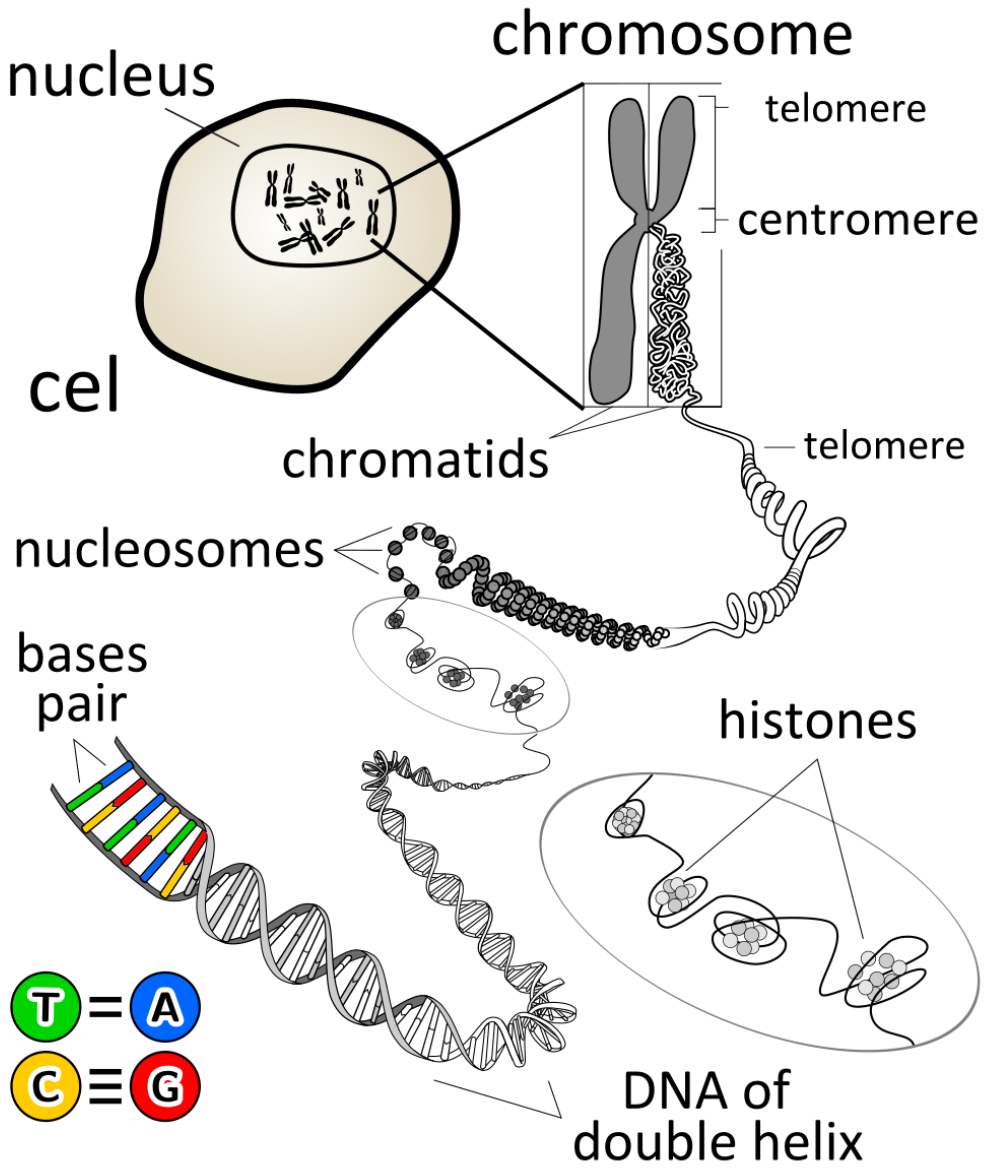
Figure 2.14 The backbone torsion angles in a unit nucleotide. Each rotatable bond is indicated by a curved arrow.

A – form: Canononical form for ds-RNA

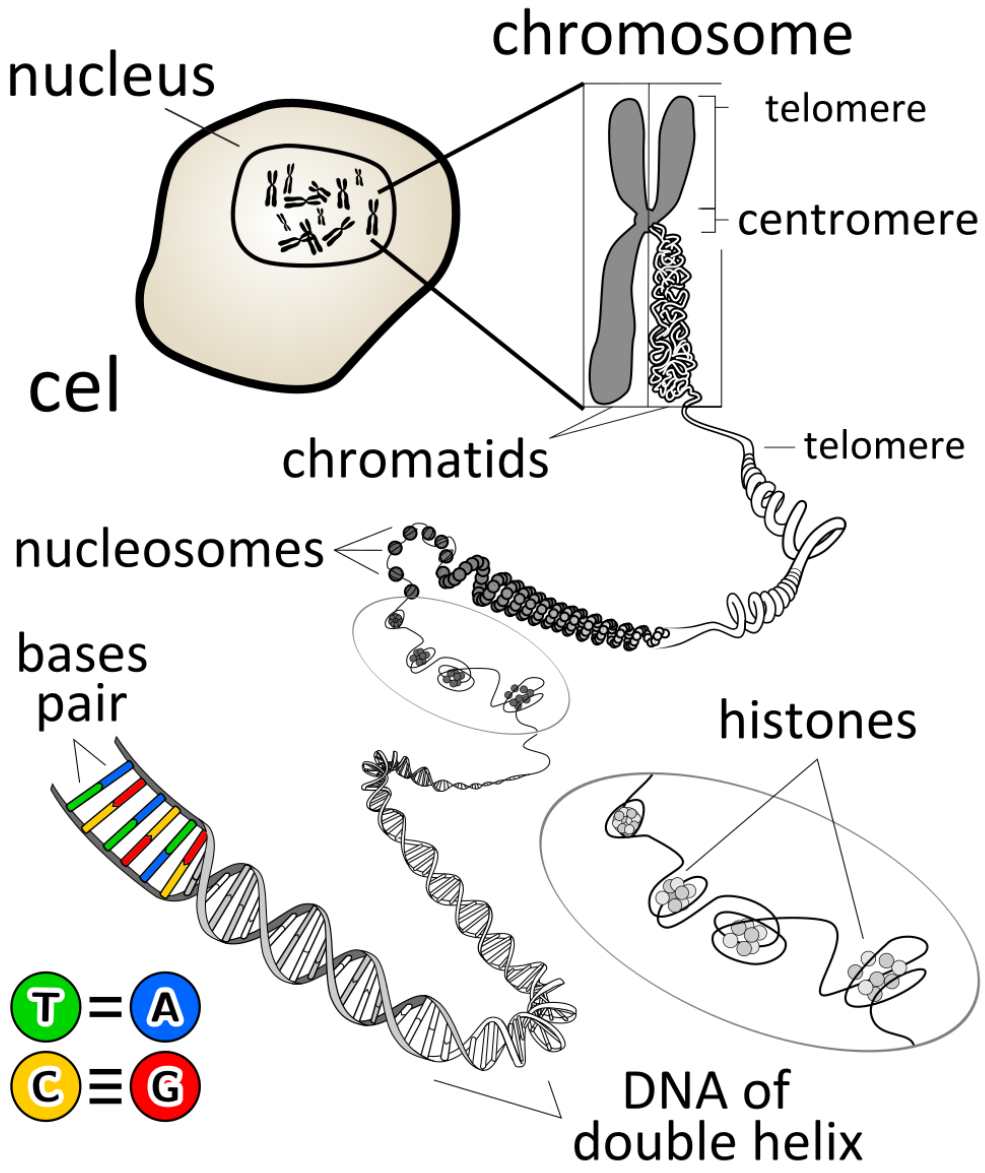


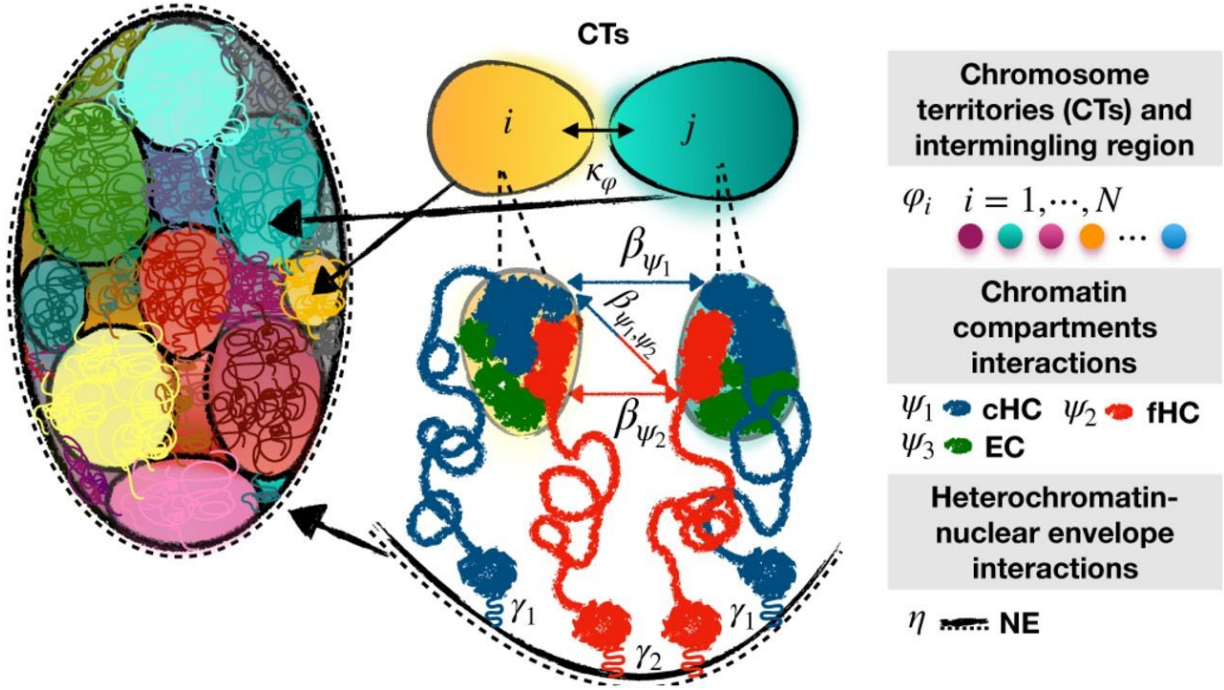
Feature	B-DNA	A-DNA	Z-DNA
Helix Type	Right-handed	Right-handed	Left-handed
Base Pairs/Turn	~10.5	~11	~12
Sugar Pucker	C2'-endo	C3'-endo	Alternating C2'-endo / C3'-endo
Groove Width	Major: Wide, Minor: Narrow	Major: Deep, Minor: Shallow	Nearly absent
Conditions	Physiological	Dehydrated	High salt, negative supercoiling

Levels of DNA organization



Levels of DNA organization

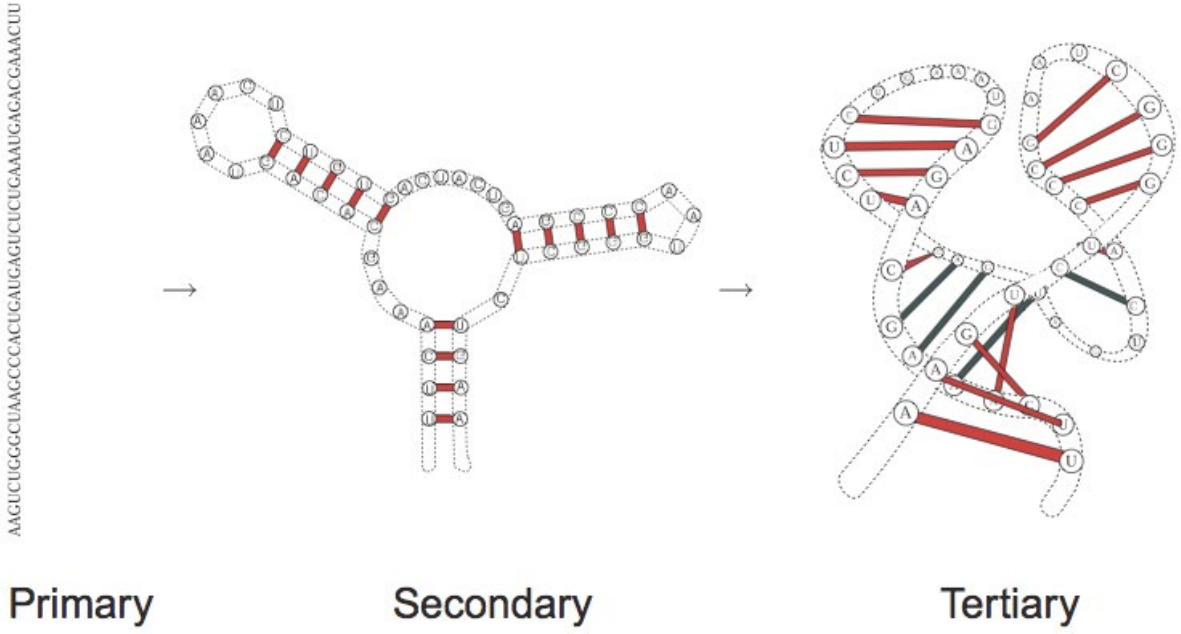




Schematic representation of the mulitphase liquid chromatin model of nucleus.

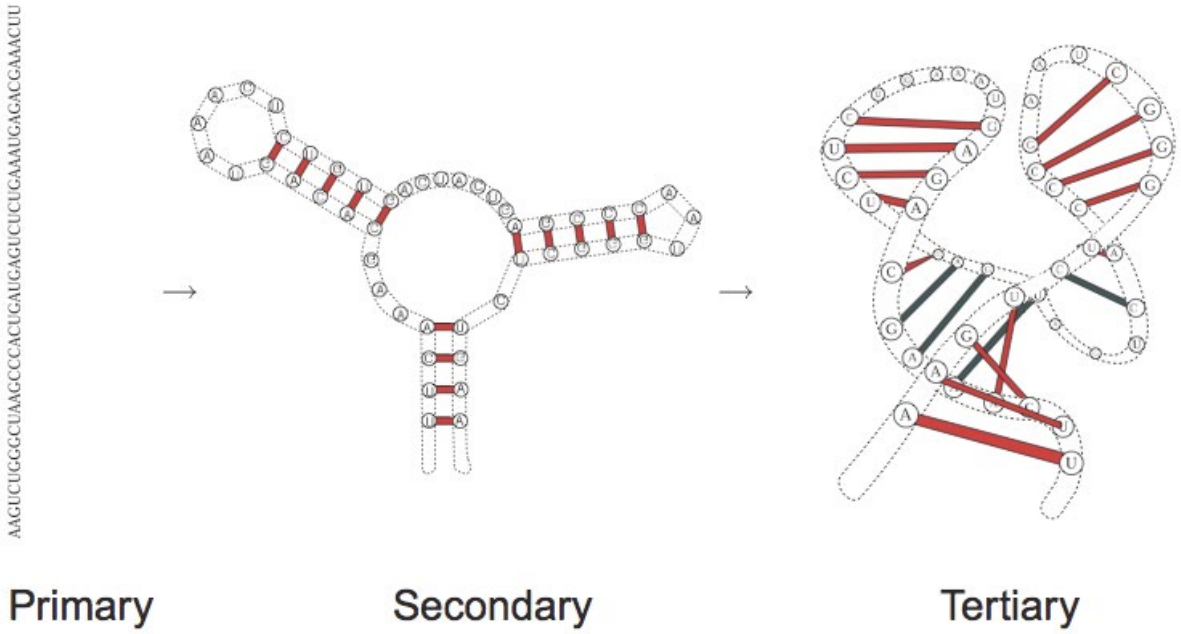
Rabia Laghmach, Michele Di Pierro, Davit A Potoyan, Biophysical Journal

RNA secondary and tertiary structure

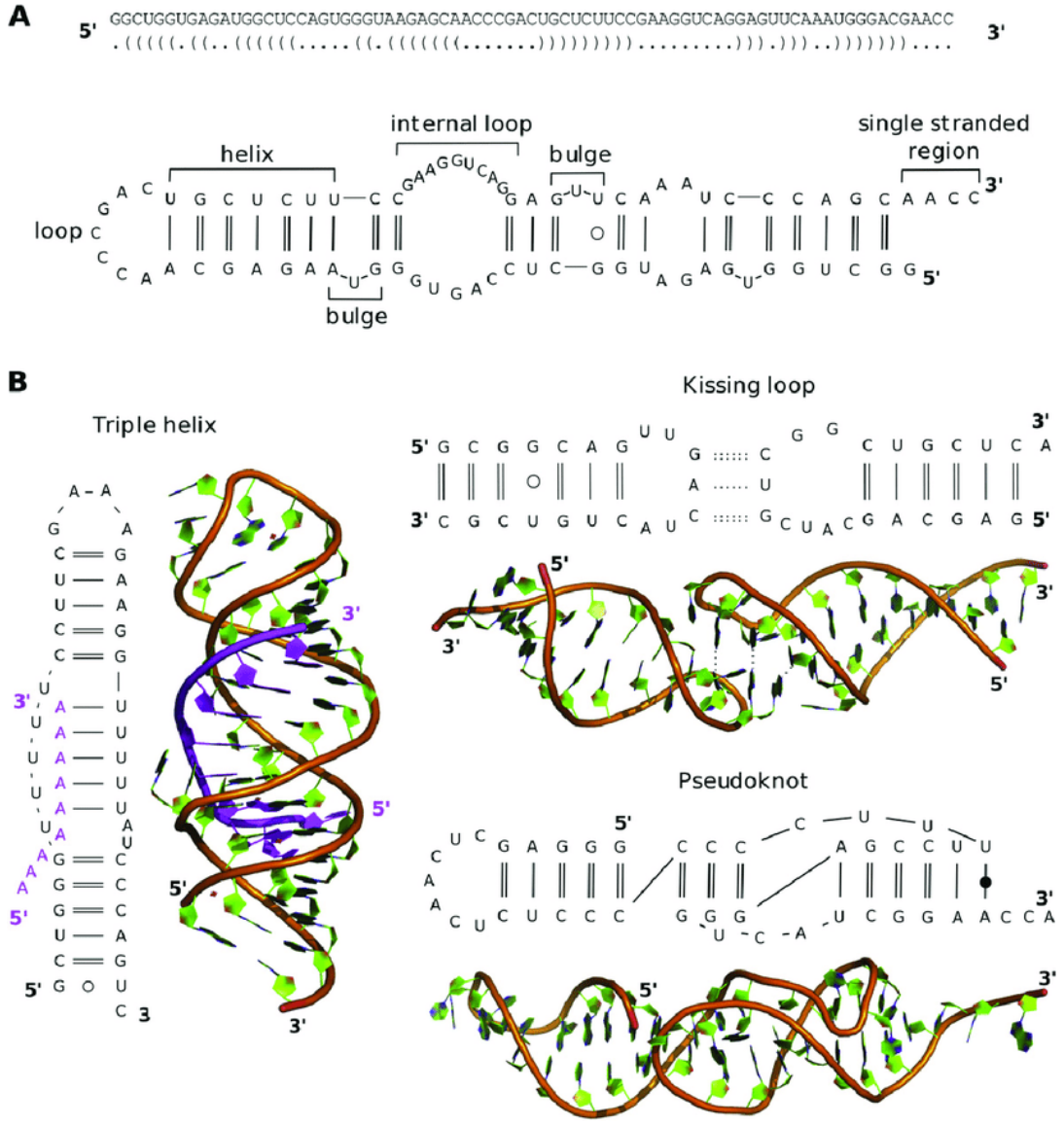


RNA is mostly a single strand molecule and can form nucleotide pairs with itself. Contrarily to DNA, RNA can also have structural roles.

RNA secondary and tertiary structure

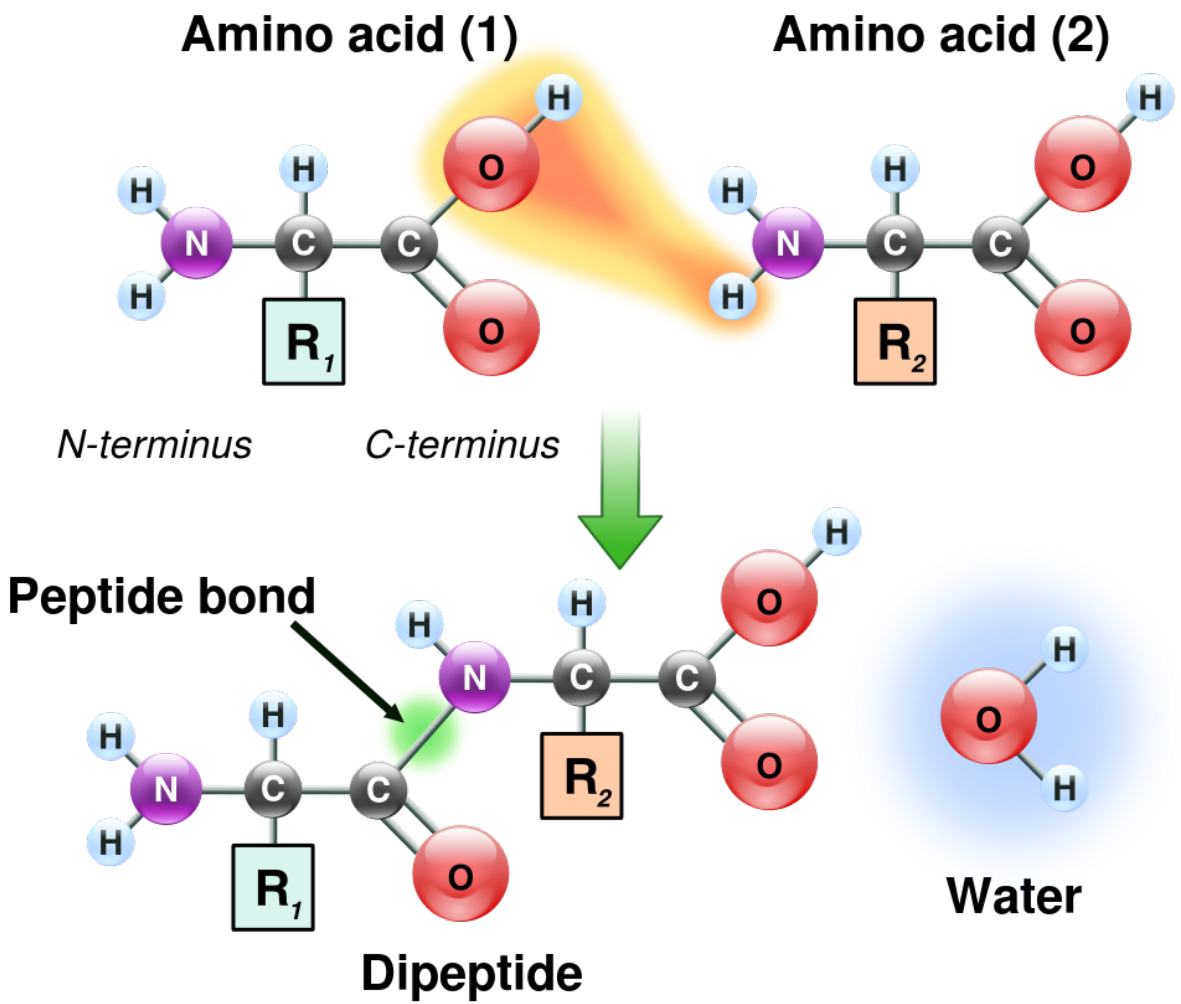


RNA is mostly a single strand molecule and can form nucleotide pairs with itself. Contrarily to DNA, RNA can also have structural roles.

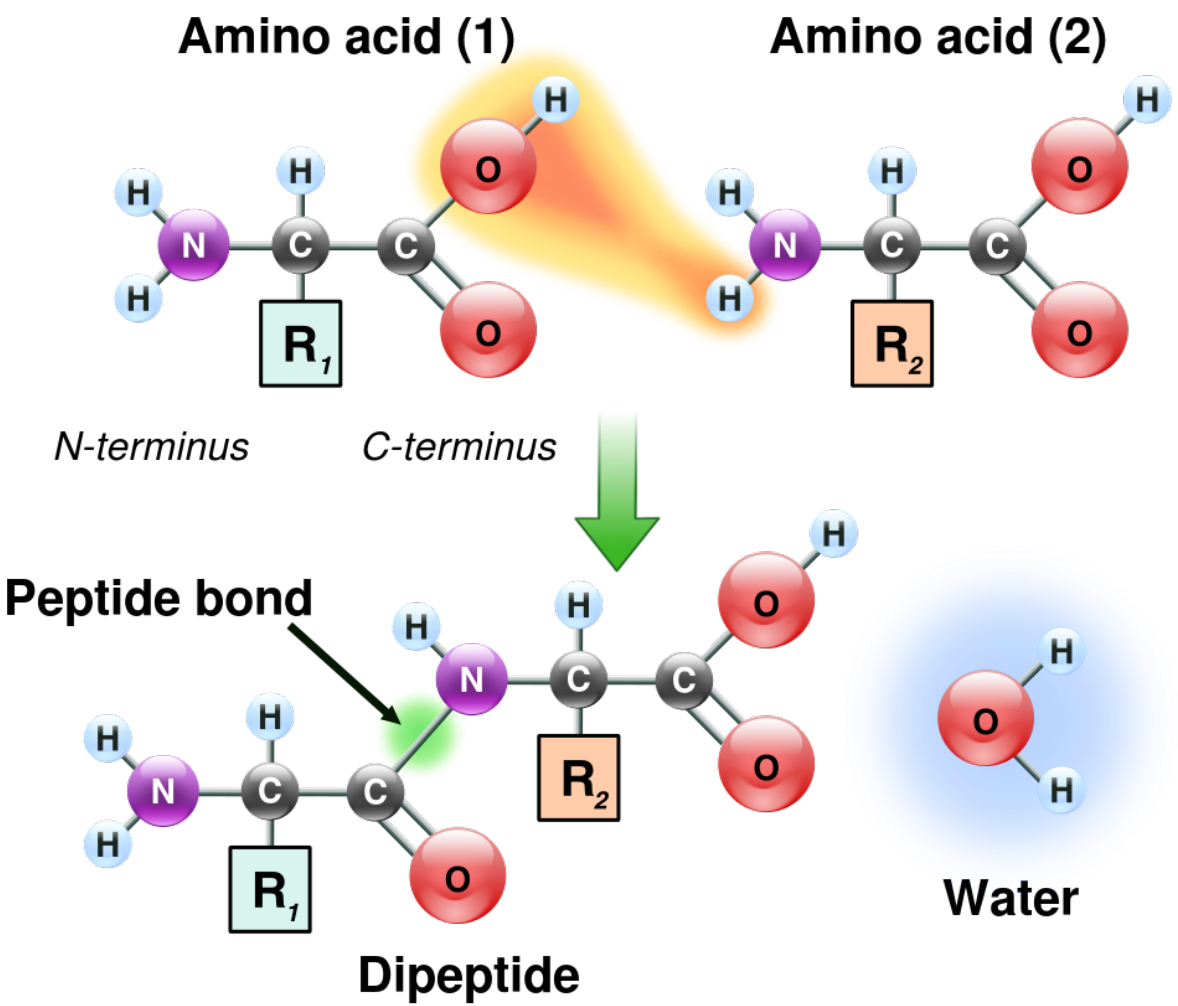


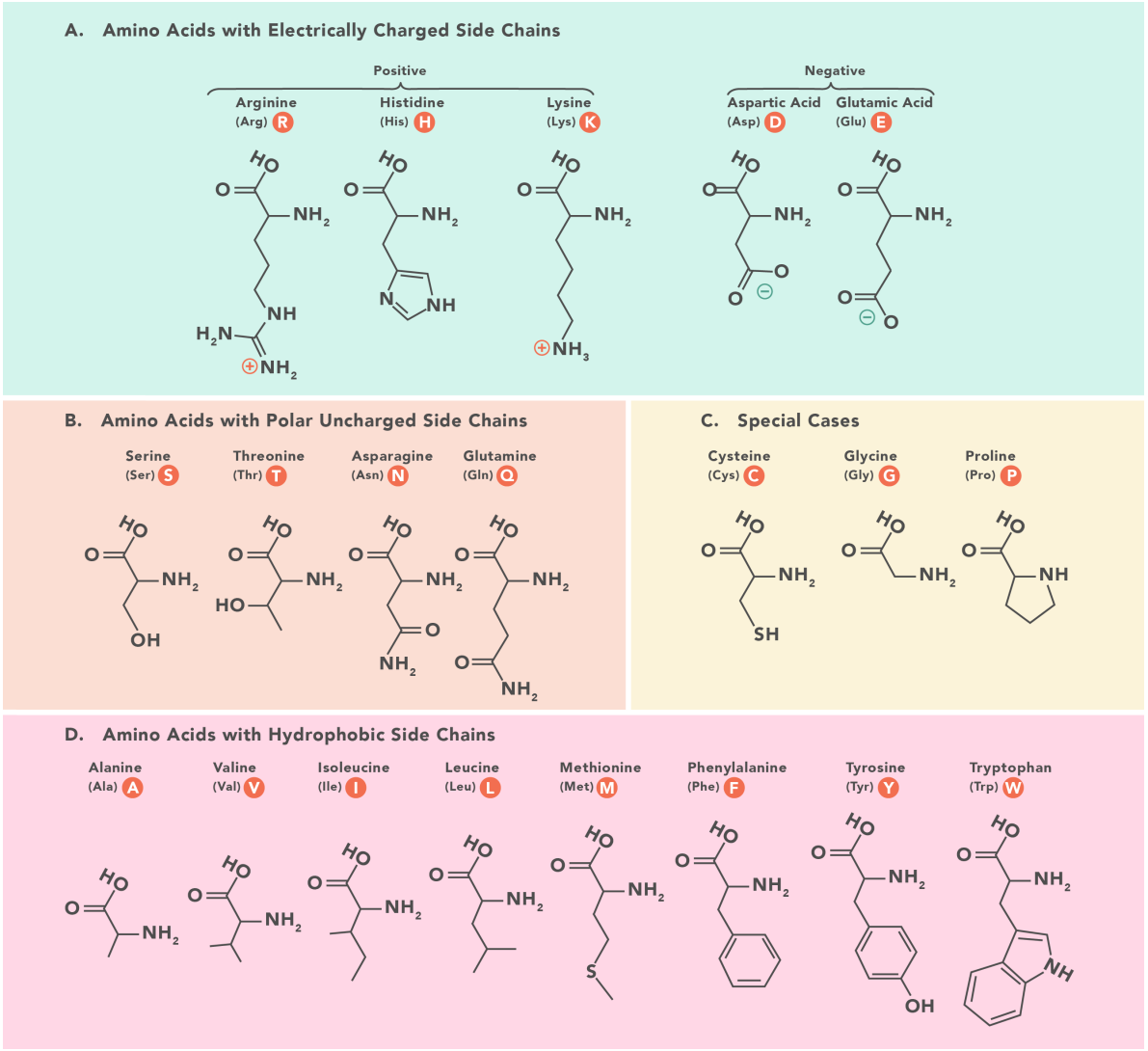
Common secondary and tertiary structures of RNA. A. The dot-bracket notation and 2-D map of a small region from the 5' end of B2-SINE lncRNA showing common secondary structure elements. B. 2-D and 3-D representations of RNA tertiary structures. (Amanda J Lewis et al. 2015)

AMINO ACIDS AND PEPTIDE BONDS

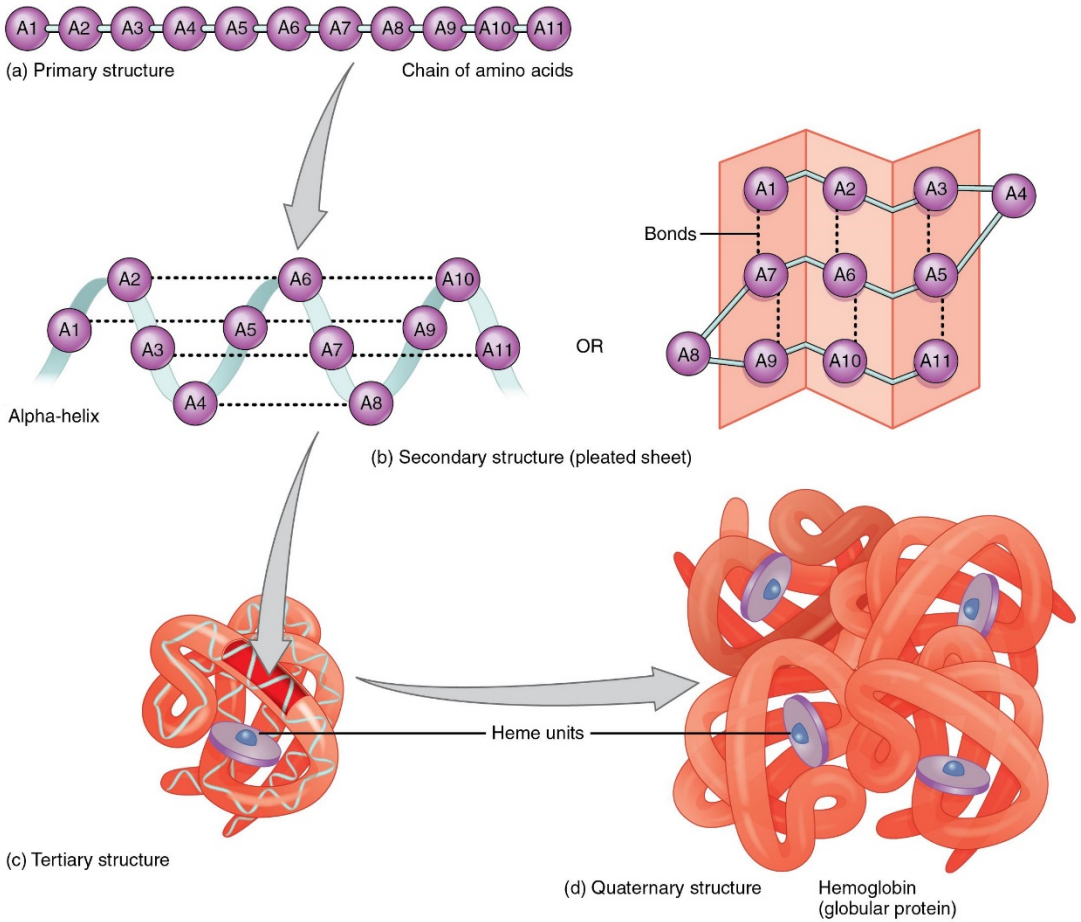


AMINO ACIDS AND PEPTIDE BONDS





PROTEIN STRUCTURE



• Primary structure: Linear sequence of amino acids linked by peptide bonds (shown as beads on a string)

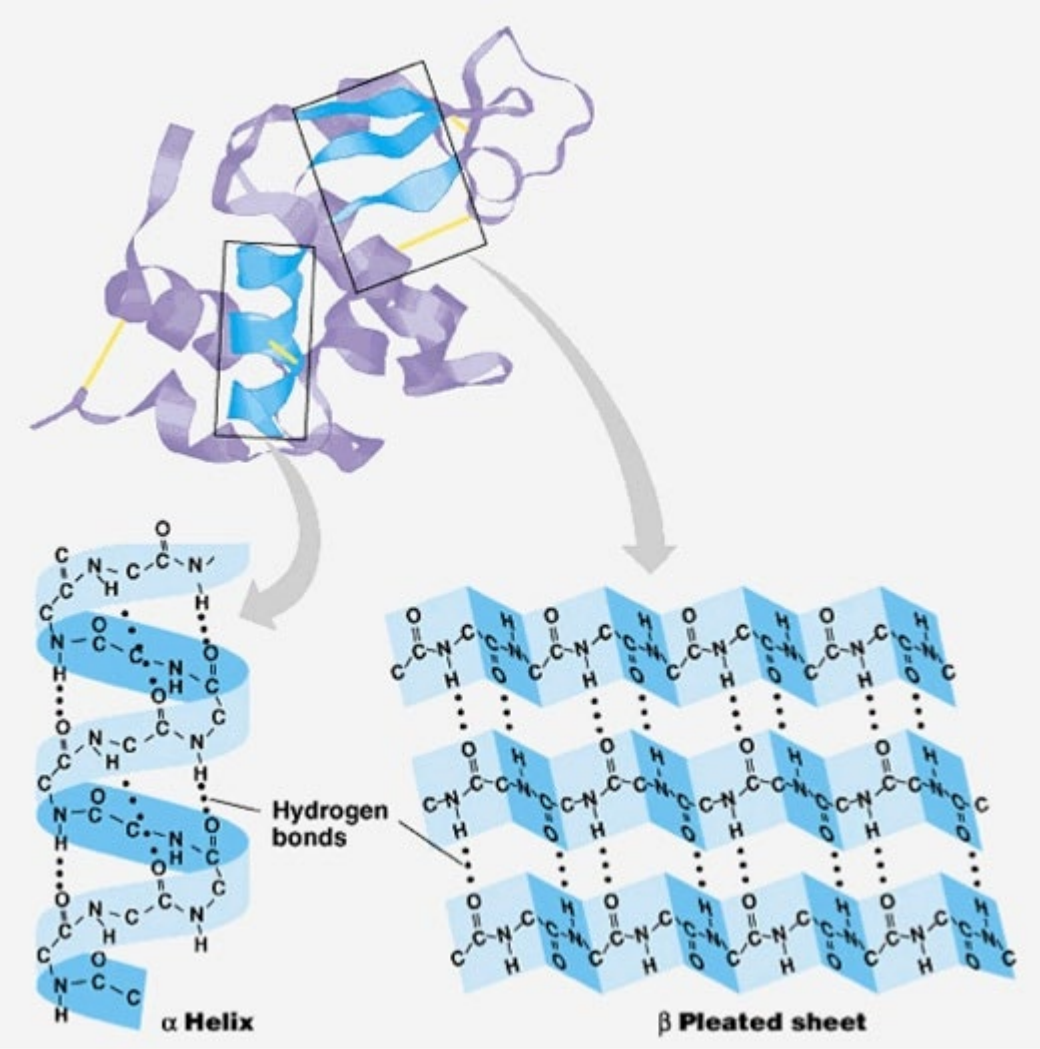
• Secondary structure: Local folding patterns stabilized by hydrogen bonds
  - Alpha helix (coiled structure)
  - Beta sheet (zigzag planar structure)

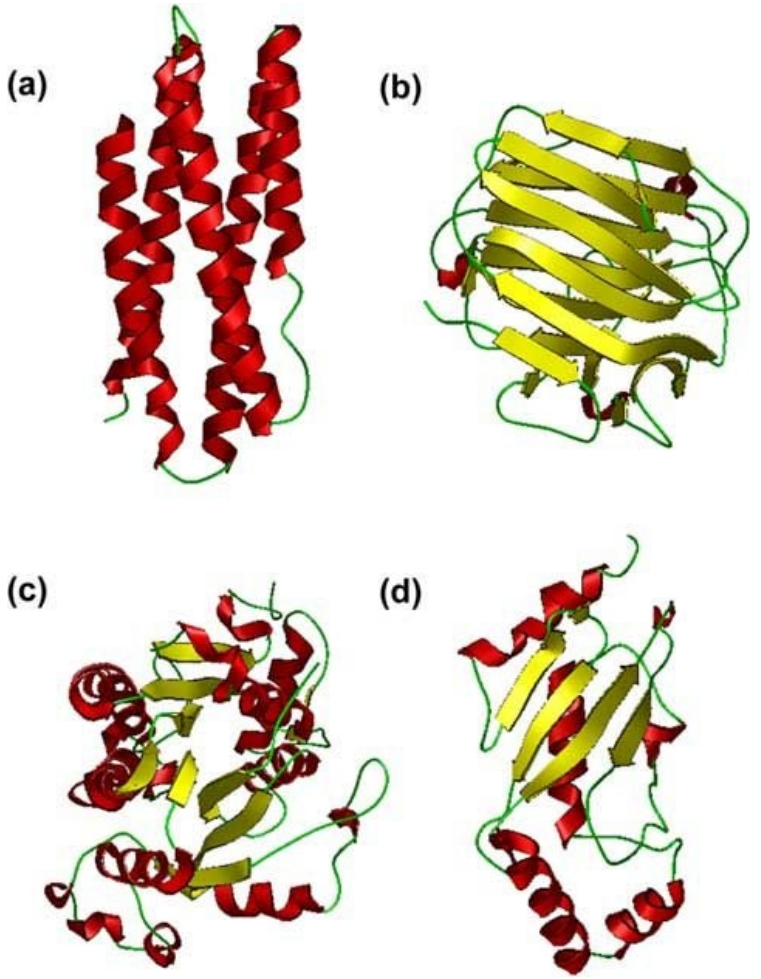
• Tertiary structure: 3D arrangement and folding of the entire polypeptide chain stabilized by various interactions (ionic, hydrogen bonds, disulfide bridges, hydrophobic)

• Quaternary structure: Complex of multiple polypeptide chains (subunits) arranged in a specific geometric pattern (e.g., hemoglobin with 4 subunits)

• Structural levels increase from primary to quaternary, with more complex folding and molecular interactions determining the final 3D shape and function

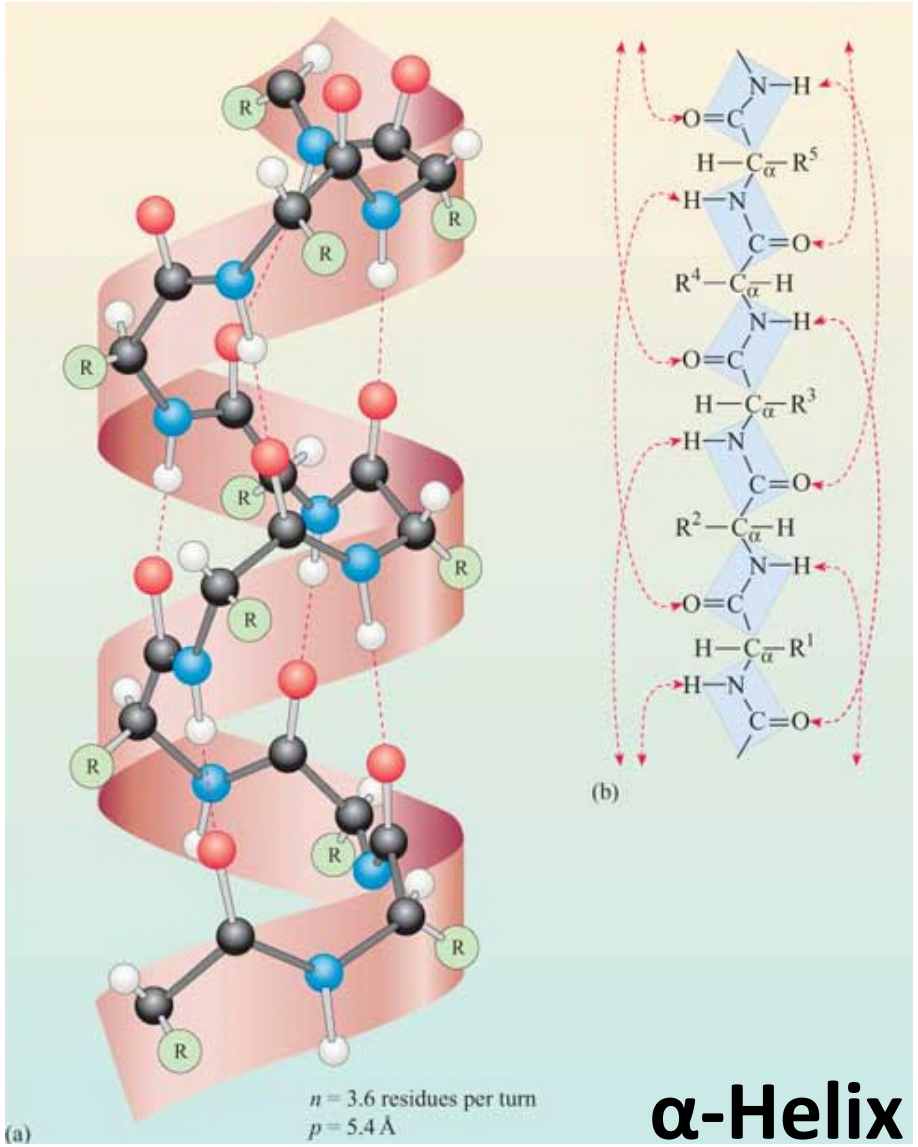
PROTEIN CLASSIFICATION BASED ON SECONDARY STRUCTURE



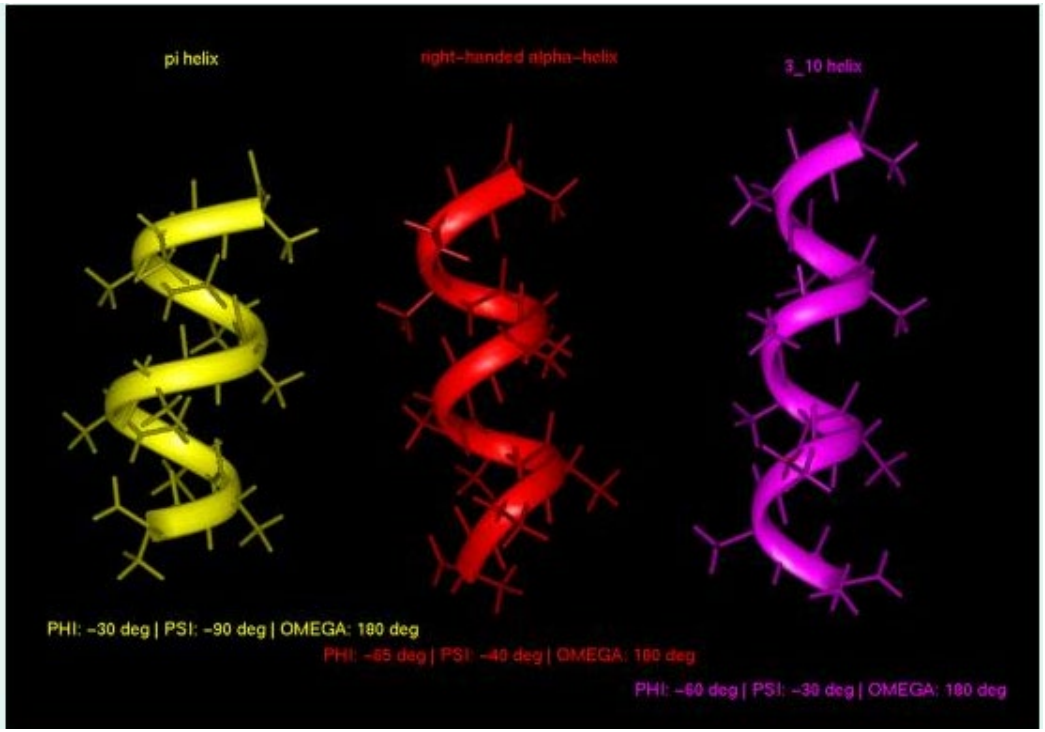


Ribbon drawings to show the four structural classes of proteins: (a) all-α, (b) all-β, (c) α/β, and (d) α+β.
Kuo-Chen Chou, 2005

HELICES

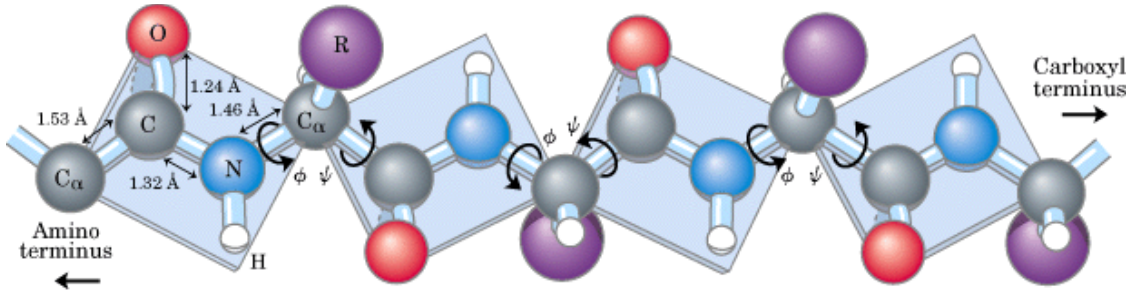


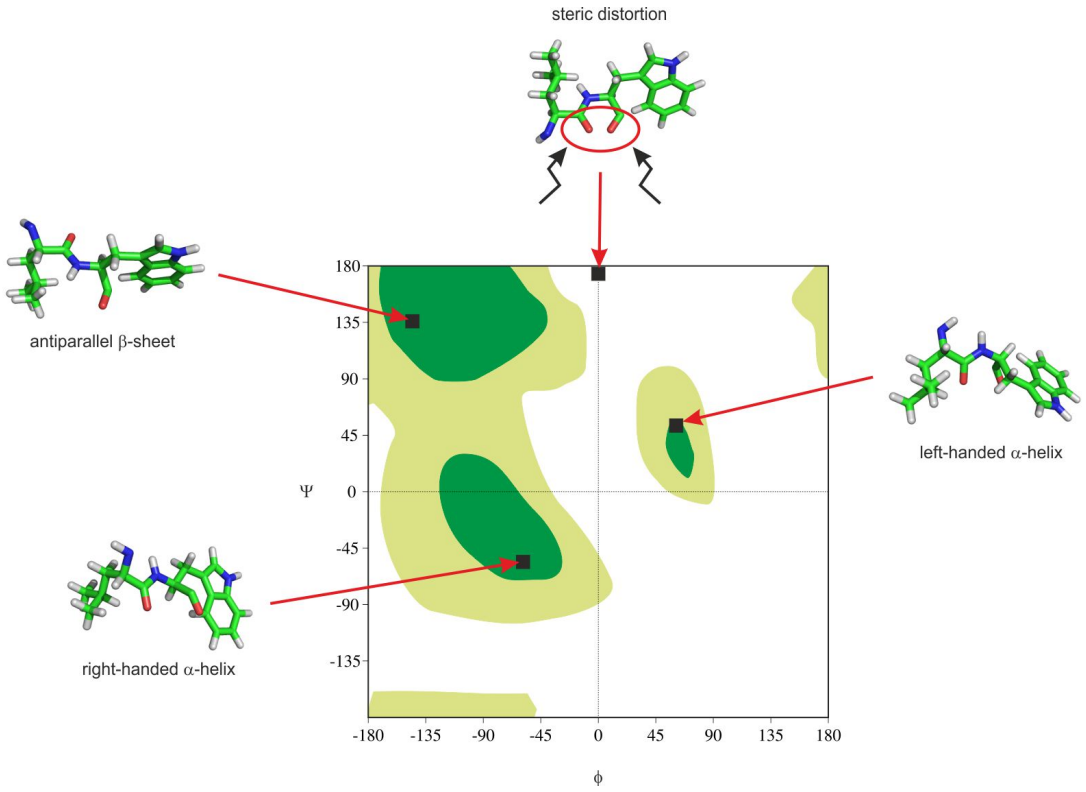
α-Helix



Feature	α-Helix	3_{10}-Helix	π-Helix
H-Bond Pattern	i → i+4	i → i+3	i → i+5
Residues per Turn	3.6	3	4.4
Helical Rise (Å/residue)	1.5	2	1.15
Pitch (Å/turn)	5.4	6	5.2
Φ (phi) angle	~ -60°	~ -49°	~ -57°
Ψ (psi) angle	~ -45°	~ -26°	~ -70°
Stability	High	Low	Low
Occurrence	Common	Short segments	Rare

Ramachandran Plots



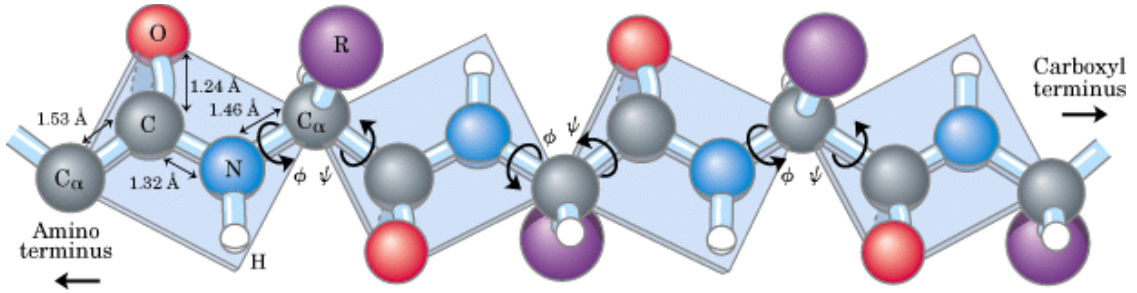


The relative position of two consecutive amino acids is described by two dihedral angles (φ and ψ).

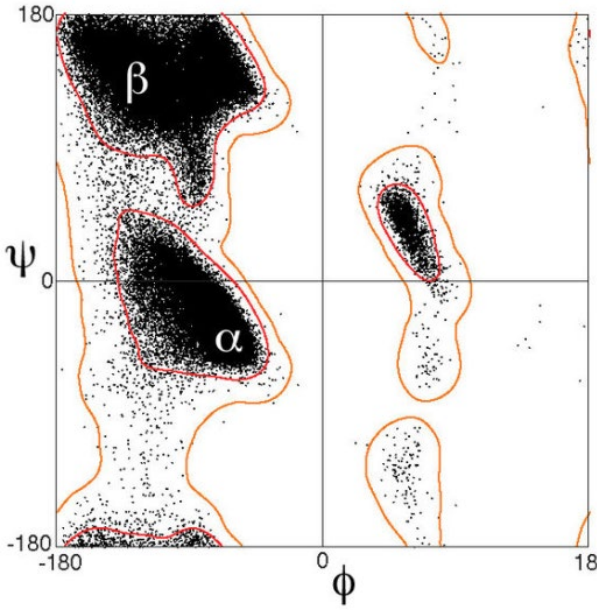
Not every combination of φ and ψ is possible, due to steric and electrostatic hindrance.

The values of the two dihedral angles is linked to the local secondary structure.

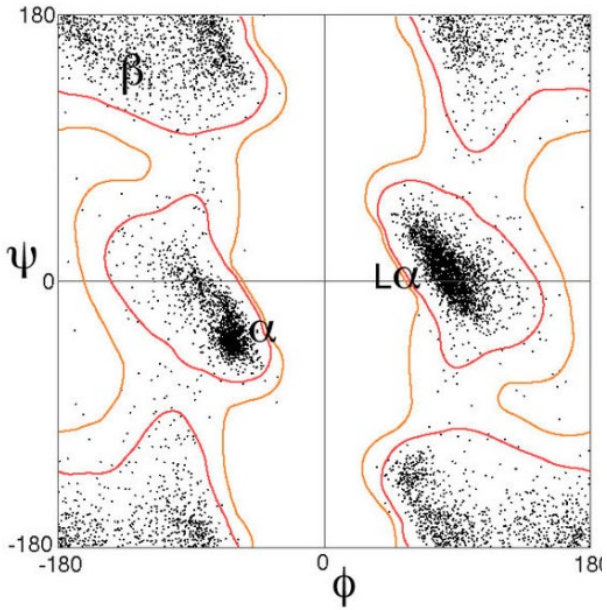
Ramachandran Plots



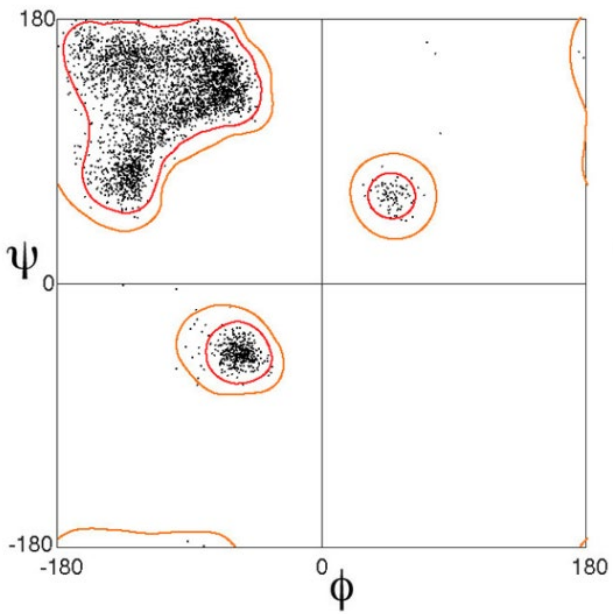
Different amino acids have different preferred orientation (which explain their secondary structure propensity).



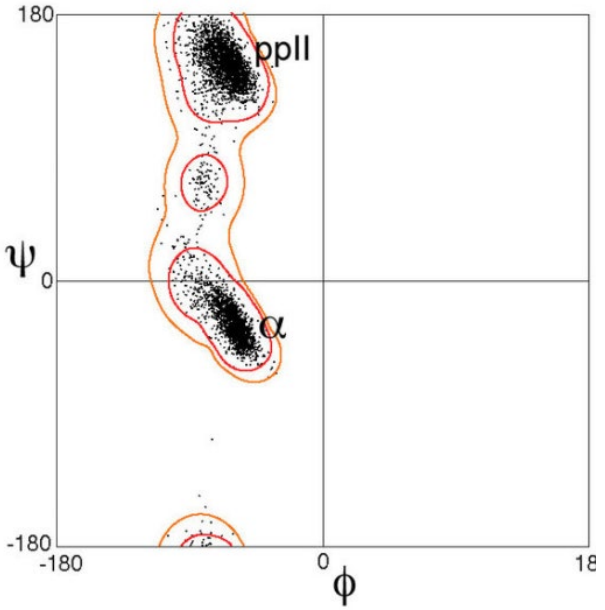
Average



Glycine



Proline



Pre-Proline

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The Protein Data Bank and PDB files




```
HEADER    TRANSFERASE                             17-JUN-02   1M17
TITLE     EPIDERMAL GROWTH FACTOR RECEPTOR TYROSINE KINASE DOMAIN
TITLE    2 WITH 4-ANILINOQUINAZOLINE INHIBITOR ERLOTINIB
COMPND    MOL_ID: 1;
COMPND   2 MOLECULE: EPIDERMAL GROWTH FACTOR RECEPTOR;
COMPND   3 CHAIN: A;
COMPND   4 FRAGMENT: TYROSINE KINASE DOMAIN (RESIDUES 671-998);
COMPND   5 SYNONYM: RECEPTOR PROTEIN-TYROSINE KINASE ERBB-1;
COMPND   6 EC: 2.7.1.112;
COMPND   7 ENGINEERED: YES
SOURCE    MOL_ID: 1;
SOURCE   2 ORGANISM_SCIENTIFIC: HOMO SAPIENS;
SOURCE   3 ORGANISM_COMMON: HUMAN;
SOURCE   4 GENE: EGFR;
SOURCE   5 EXPRESSION_SYSTEM: SPODOPTERA FRUGIPERDA;
SOURCE   6 EXPRESSION_SYSTEM_COMMON: FALL ARMYWORM;
SOURCE   7 EXPRESSION_SYSTEM_STRAIN: AUTOGRAPHICA
SOURCE   8 CALIFORNICA/T.NICOPLUSIA;
SOURCE   9 EXPRESSION_SYSTEM_CELL_LINE: SF9;
SOURCE  10 EXPRESSION_SYSTEM_VECTOR_TYPE: PLASMID;
SOURCE  11 EXPRESSION_SYSTEM_PLASMID: PVL1392
KEYWDS    TRANSFERASE, TYROSINE KINASE DOMAIN
EXPDTA    X-RAY DIFFRACTION
AUTHOR    J.STAMOS,M.X.SLIWKOWSKI,C.EIGENBROT
REVDAT   2   25-FEB-03 1M17    1       JRNL
REVDAT   1   04-SEP-02 1M17    0
JRNL        AUTH   J.STAMOS,M.X.SLIWKOWSKI,C.EIGENBROT
JRNL        TITL   STRUCTURE OF THE EPIDERMAL GROWTH FACTOR RECEPTOR
JRNL        TITL 2 KINASE DOMAIN ALONE AND IN COMPLEX WITH A
JRNL        TITL 3 4-ANILINOQUINAZOLINE INHIBITOR.
JRNL        REF    J.BIOL.CHEM.                  V. 277 46265 2002
JRNL        REFN   ASTM JBCHA3  US ISSN 0021-9258
REMARK   1
REMARK   2
REMARK   2 RESOLUTION. 2.60 ANGSTROMS.
```

PDB: Protein Data Bank *https://www.rcsb.org/*

The Protein Data Bank and PDB files

Atom number, Atom "Type", Residue Name, Chain, Residue Number

Atom coordinates (Å)

Occupancy (fraction of time spent in that position for each atom)

B-factors (fluctuations around the average positions Å^2)

Everything following the "ATOM" word indicates the properties of an atom in the protein

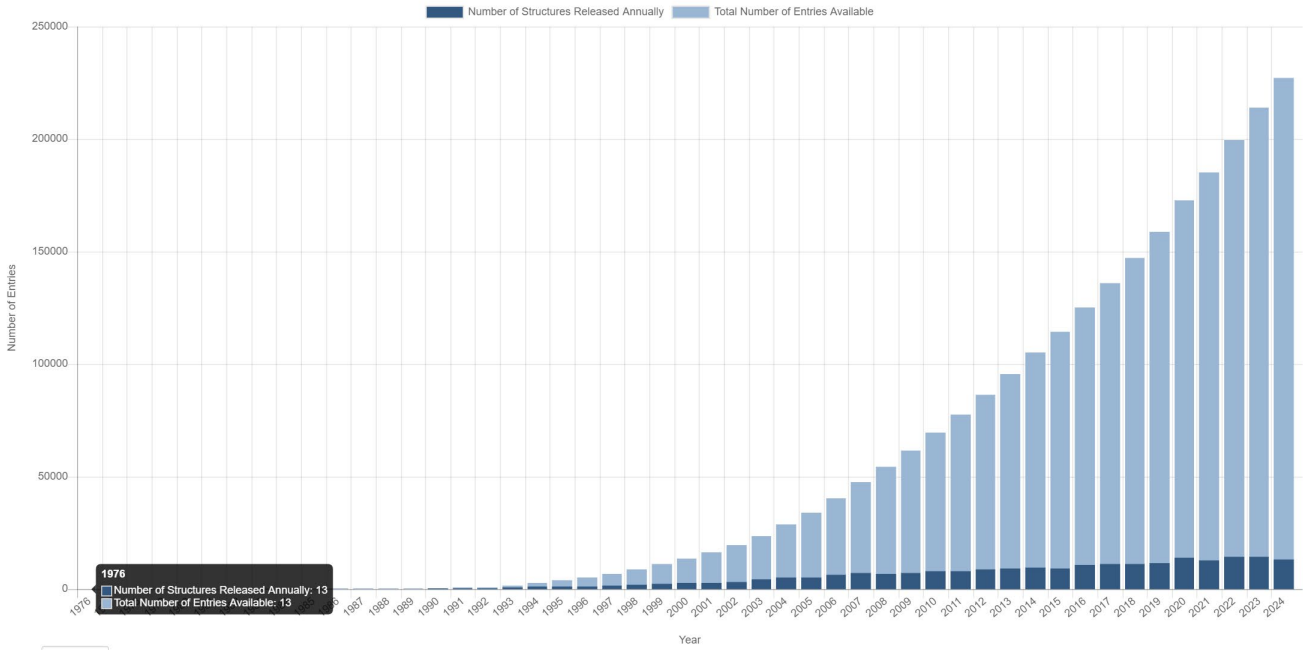
Element

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ATOM      2  CA  ARG A  20      24.786 -38.383  -3.724  1.00 48.20           C
ATOM      3  C   ARG A  20      26.209 -38.780  -4.233  1.00 46.86           C
ATOM      4  O   ARG A  20      26.312 -39.492  -5.209  1.00 46.96           O
ATOM      5  CB  ARG A  20      24.328 -37.035  -4.295  1.00 48.44           C
ATOM      6  CG  ARG A  20      23.791 -37.076  -5.718  1.00 49.20           C
ATOM      7  CD  ARG A  20      23.543 -35.655  -6.309  1.00 49.60           C
ATOM      8  NE  ARG A  20      22.419 -34.974  -5.669  1.00 54.70           N
ATOM      9  CZ  ARG A  20      22.453 -33.754  -5.107  1.00 55.47           C
ATOM     10  NH1 ARG A  20      23.558 -33.001  -5.109  1.00 52.88           N
ATOM     11  NH2 ARG A  20      21.345 -33.268  -4.550  1.00 55.28           N
ATOM     12  N   SER A  21      27.281 -38.306  -3.583  1.00 45.68           N
ATOM     13  CA  SER A  21      28.663 -38.666  -3.946  1.00 43.64           C
ATOM     14  C   SER A  21      29.113 -39.741  -2.991  1.00 42.98           C
ATOM     15  O   SER A  21      28.807 -39.631  -1.810  1.00 43.65           O
ATOM     16  CB  SER A  21      29.618 -37.466  -3.846  1.00 43.49           C
ATOM     17  OG  SER A  21      29.413 -36.523  -4.890  1.00 41.49           O
ATOM     18  N   PHE A  22      29.822 -40.764  -3.483  1.00 41.35           N
ATOM     19  CA  PHE A  22      30.314 -41.886  -2.644  1.00 40.87           C
ATOM     20  C   PHE A  22      31.844 -42.145  -2.634  1.00 39.14           C
ATOM     21  O   PHE A  22      32.290 -43.097  -1.991  1.00 39.49           O
ATOM     22  CB  PHE A  22      29.550 -43.211  -2.938  1.00 41.87           C
ATOM     23  CG  PHE A  22      28.062 -43.084  -2.760  1.00 44.68           C
ATOM     24  CD1 PHE A  22      27.255 -42.664  -3.815  1.00 46.93           C
ATOM     25  CD2 PHE A  22      27.468 -43.322  -1.516  1.00 49.07           C
ATOM     26  CE1 PHE A  22      25.864 -42.500  -3.642  1.00 50.21           C
ATOM     27  CE2 PHE A  22      26.069 -43.156  -1.328  1.00 50.25           C
ATOM     28  CZ  PHE A  22      25.272 -42.759  -2.394  1.00 48.69           C
```

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The Protein Data Bank and PDB files

PDB Statistics: Overall Growth of Released Structures Per Year



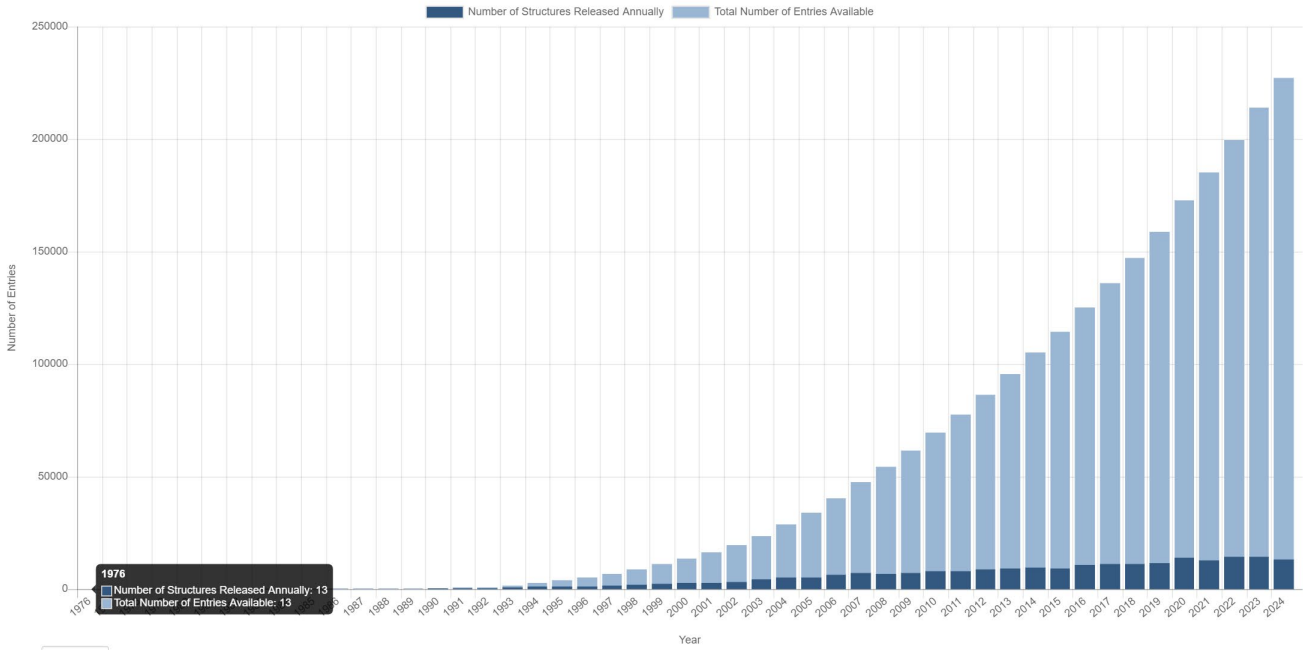
Show 10 entries

Copy CSV

Year	Total Number of Entries Available	Number of Structures Released Annually
2024	227,111	13,038
2023	214,073	14,464
2022	199,609	14,258
2021	185,351	12,571
2020	172,780	13,985
2019	158,795	11,472
2018	147,323	11,155
2017	136,168	11,059
2016	125,109	10,805
2015	114,304	9,244

The Protein Data Bank and PDB files

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PDB Data Distribution by Experimental Method and Molecular Type

All Statistics

Copy CSV

Molecular Type	X-ray	EM	NMR	Multiple methods	Neutron	Other	Total
Protein (only)	167,566	15,792	12,538	208	79	32	196,215
Protein/Oligosaccharide	9,658	2,652	34	8	2	0	12,354
Protein/NA	8,743	4,736	286	7	0	0	13,772
Nucleic acid (only)	2,869	138	1,510	14	3	1	4,535
Other	170	10	33	0	0	0	213
Oligosaccharide (only)	11	0	6	1	0	4	22
Total	189,017	23,328	14,407	238	84	37	227,111

PDB Structures with Experimental Data

179,097 have a structure factor file
11,505 have an NMR restraint file
6,038 have a chemical shifts file
4,212 have NMR unified data files (NEF and/or NMR-STAR format files)
22,829 3DEM maps were used for the determination of 23,056 PDB structures

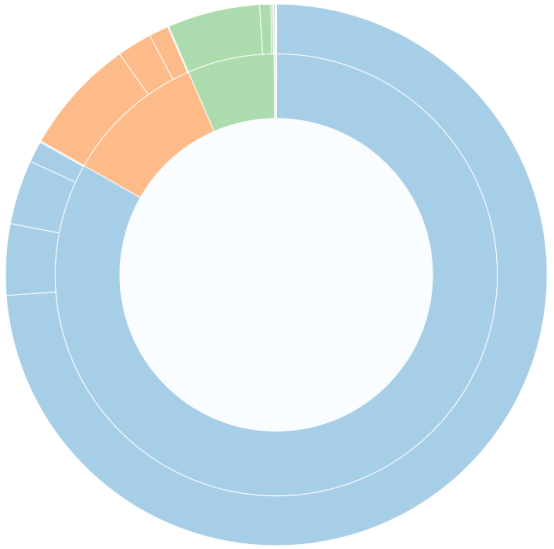
PDB Structures with Small Molecule Data

172,253 with any non-polymer small molecules
170,696 with non-polymer small molecules of 100-300 Da
59,444 with non-polymer small molecules of 300-500 Da
40,786 with non-polymer small molecules of 500-1200 Da
38,456 with author-designated Ligand/s of Interest
4,537 with a biologically interesting short oligomer or oligomer-like BIRD molecules
12,534 with with branched entities of oligosaccharides

Sunburst Chart for Experimental Method and Molecular Type

Inner most layer represents the distribution by experimental methods, X-ray is shown in blue; NMR in orange; EM in green; Multiple methods in yellow; Neutron in pink; Other in grey. Outer most layer further shows the distribution by molecular type of each correspond experimental method.

Mouse over to view the statistical description of the particular section; click on a section to zoom in that particular section's statistical distribution, and then click on the center bullseye to zoom out back to the previous distribution.

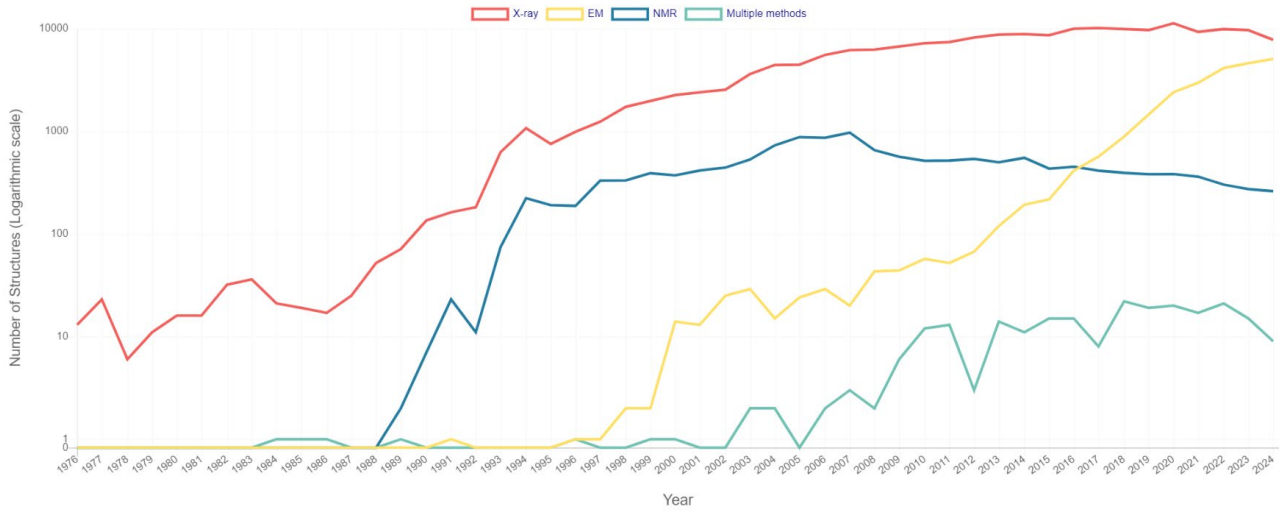


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The Protein Data Bank and PDB files

All Statistics

Number of Released PDB Structures per Year



- X-ray: X-RAY DIFFRACTION, FIBER DIFFRACTION, or POWDER DIFFRACTION
- NMR: SOLUTION NMR or SOLID-STATE NMR
- EM: ELECTRON MICROSCOPY or ELECTRON CRYSTALLOGRAPHY or ELECTRON TOMOGRAPH
- MULTIPLE METHODS: Multiple experimental methods. For example, if a structure is solved by X-RAY DIFFRACTION AND NEUTRON DIFFRACTION, it will be counted only in this category.

Show 10 entries

Copy CSV

Year	X-Ray	NMR	EM	Multiple Methods
2024	7,731	260	5,030	9
2023	9,597	272	4,579	15
2022	9,827	301	4,106	21
2021	9,240	360	2,951	17
2020	11,196	381	2,387	20
2019	9,620	380	1,451	19
2018	9,853	392	882	22
2017	10,069	412	564	8
2016	9,925	449	412	15
2015	8,578	431	216	15

All Statistics

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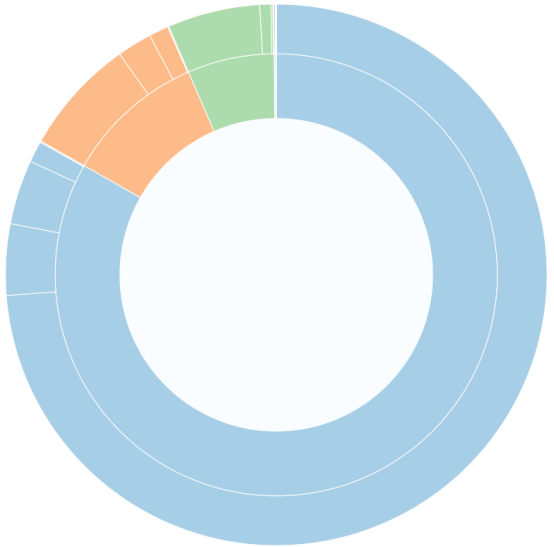
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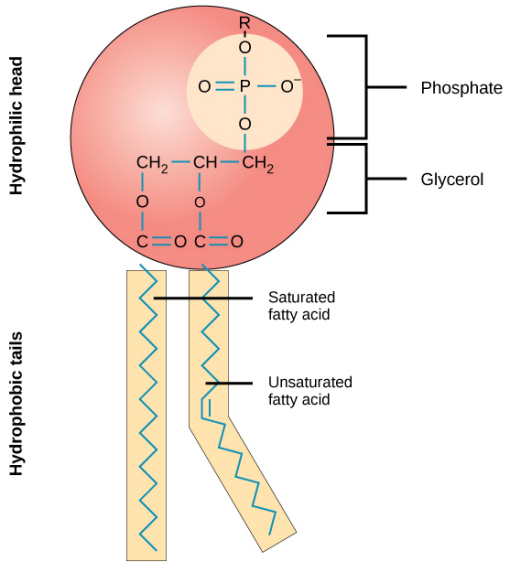
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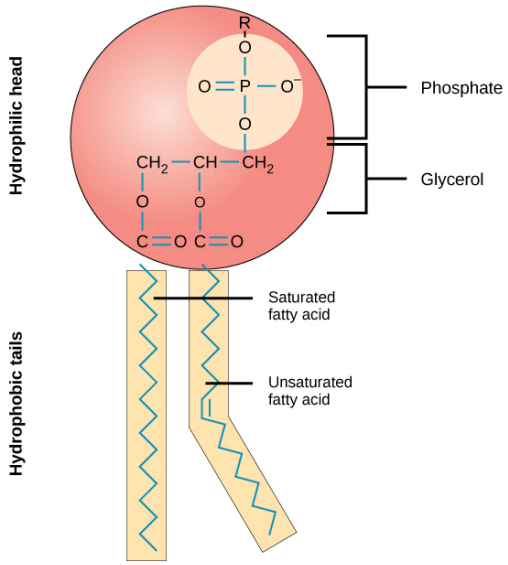
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Phospholipids and membranes

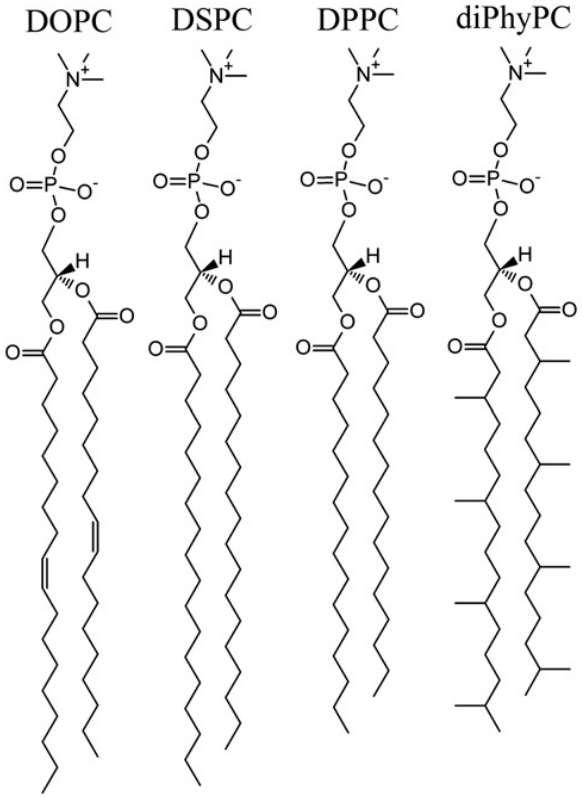


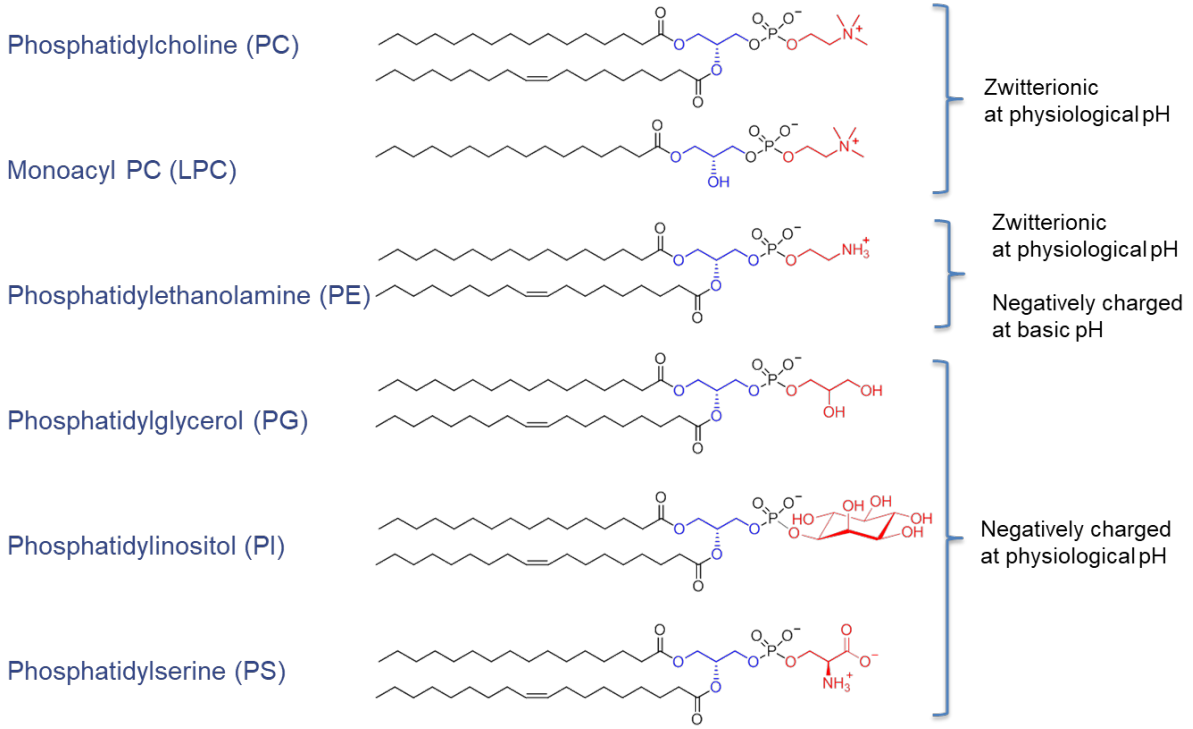
Phospholipids are amphipathic molecules: partially hydrophobic and partially hydrophilic

Phospholipids and membranes



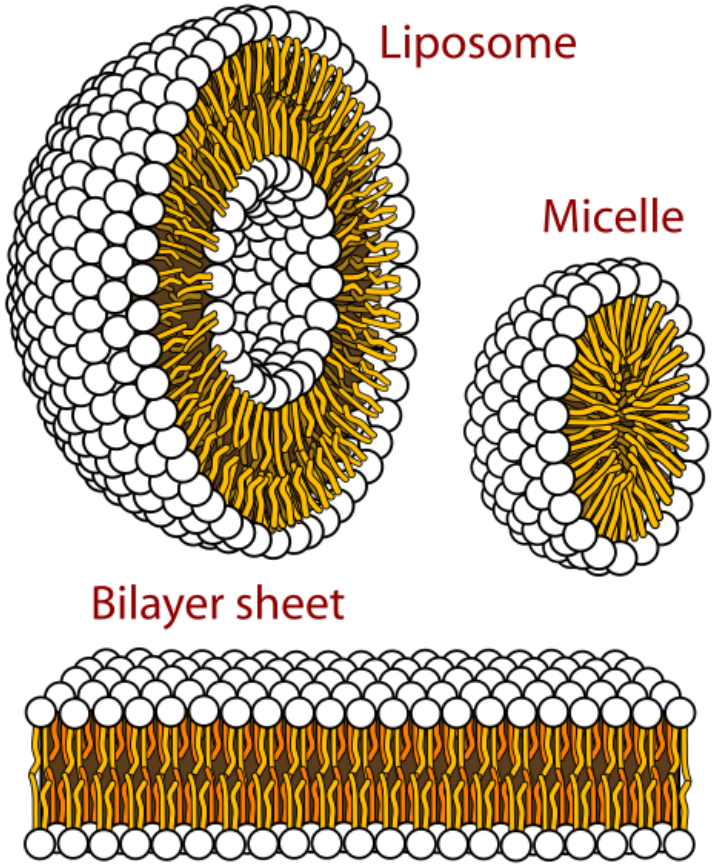
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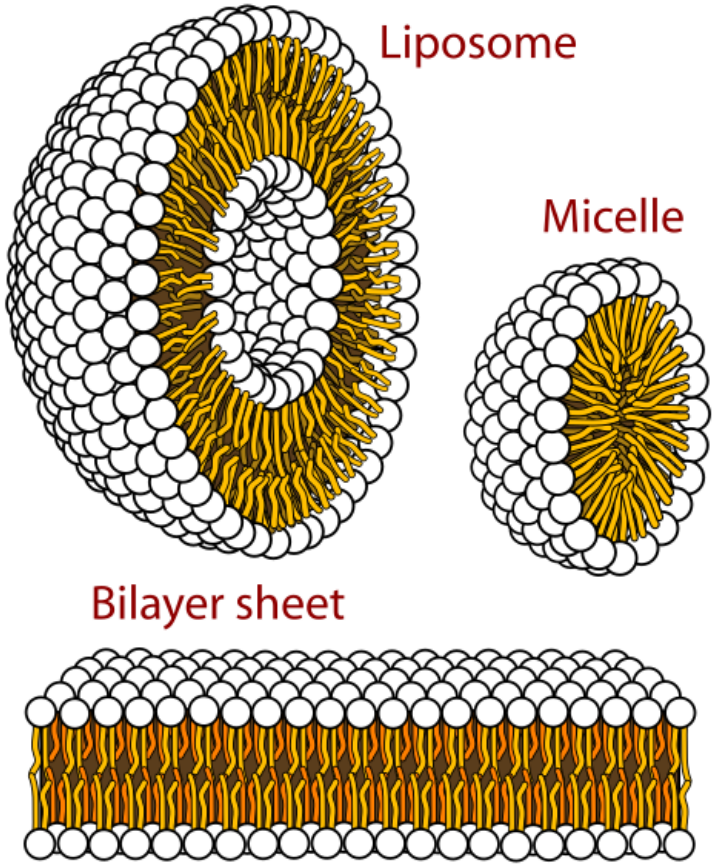
Phospholipds can be composed of various lipidic tails and different hydrophobic head (alcohol bound to the phosphate group).

Phospholipids and membranes

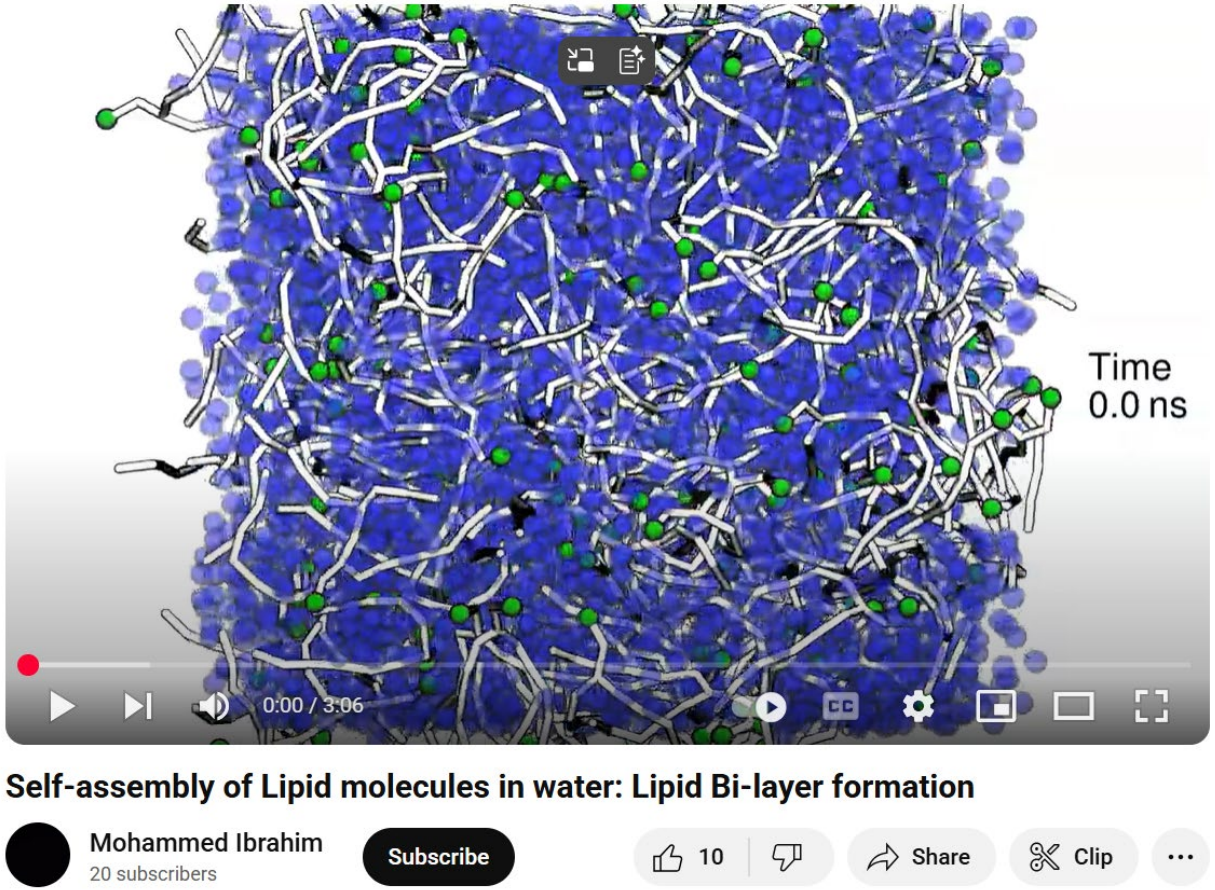


Due to their amphipathic nature, phospholipids self assemble in structure which expose the head to the water and keep the tails far from the water.

Phospholipids and membranes



Due to their amphipathic nature, phospholipids self assemble in structure which expose the head to the water and keep the tails far from the water.



[Self-assembly of Lipid molecules in water: Lipid Bi-layer formation]

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